

Excel Practical Assignment

📝 **How to use this page:** Each grey box holds the answer. Paste your screenshot under the word "Screenshot:" inside the box.
Dataset: Sample - Superstore (Tableau community link in the assignment PDF)

DASHBOARD

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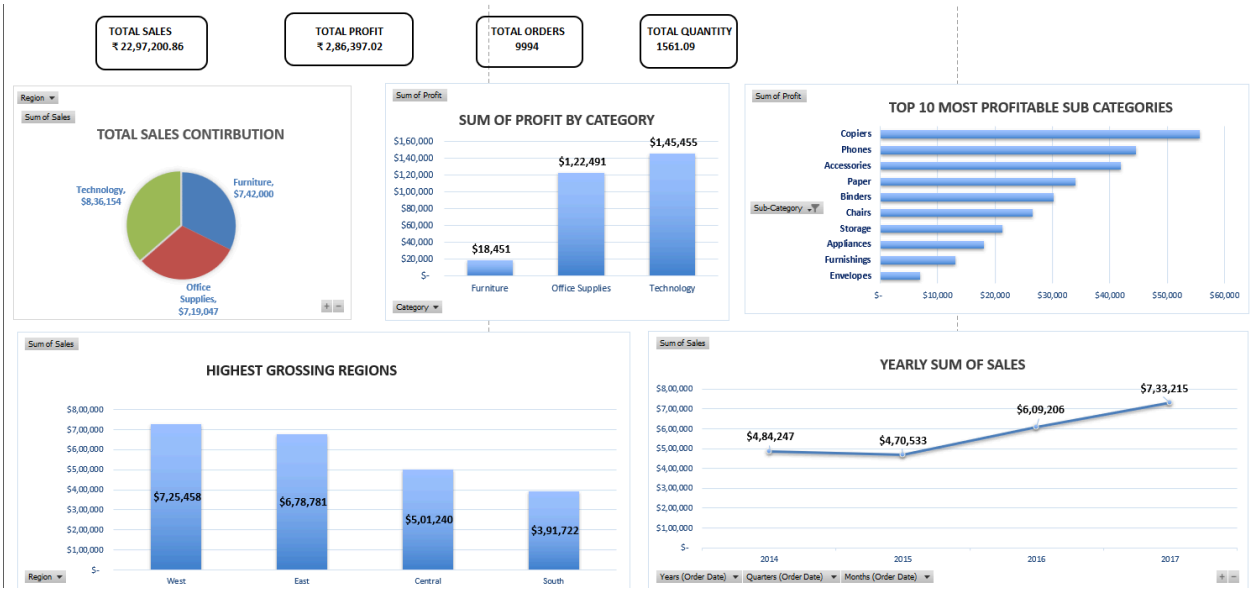
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DASHBOARD



Section 1: Basics of Excel

1.1 Basic Arithmetic Operations (SUM, MAX, MIN, SMALL, LARGE & MEDIAN)

Task 1: Create a dataset of sales figures for the year (monthly data). Calculate the total sales using the SUM function.

📸 **Answer:** Monthly sales Jan to Dec (12,000; 15,500; 9,800; 22,000; 18,400; 14,100; 27,500; 19,500; 13,650; 15,100; 14,000; 17,650). `=SUM(B2:B13)` gives **₹1,99,200**.

Task 2: Determine the highest sales month using the MAX function.

📸 **Answer:** `=MAX(B2:B13)` gives **₹27,500**, which is July.

Task 3: Identify the lowest sales month using the MIN function.

📸 **Answer:** `=MIN(B2:B13)` gives **₹9,800**, which is March.

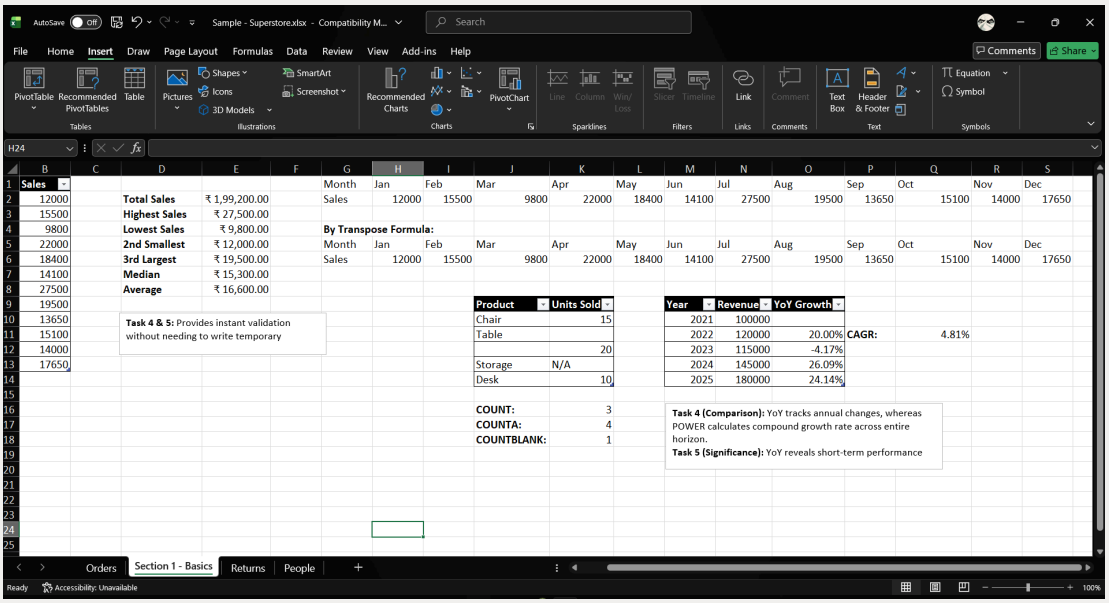
Task 4: Use the SMALL function to find the month with the 2nd lowest sales.

📸 **Answer:** `=SMALL(B2:B13,2)` gives **₹12,000**, which is January.

Task 5: Apply the LARGE function to find the month with the 3rd highest sales and calculate the median sales using the MEDIAN function.

📸 **Answer:** `=LARGE(B2:B13,3)` gives **₹19,500**, which is August. `=MEDIAN(B2:B13)` gives **₹15,300**.

Screenshot:



1.2 Transpose Vs TRANSPOSE Formula

Task 1: Create a table of monthly sales data (months as rows, sales as columns) and manually transpose it.



Answer: Copy the vertical table, then use Paste Special → Transpose. Months now run across a row (Jan to Dec) with sales beneath each month.

Task 2: Use the TRANSPOSE function to transpose the same table.



Answer: Select a horizontal range, type `=TRANSPOSE(A2:B13)` and confirm. This is the "By Transpose Formula" table.

Task 3: Compare the manual transpose and the TRANSPOSE function results.

📸 **Answer:** Both give identical values for all 12 months. The difference is that the manual result is static while the formula result is linked to the source.

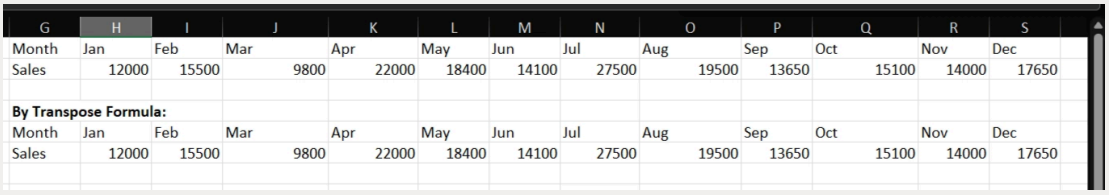
Task 4: Explain the advantages of using the TRANSPOSE function over manual transposing.

📸 **Answer:** TRANSPOSE stays linked to the source, so it updates automatically. It needs no repeat copy and paste, and it avoids stale data and human error.

Task 5: Create a dynamic range that updates the transposed values automatically when the original data changes.

📸 **Answer:** The TRANSPOSE formula range is dynamic. Change a source sales value (for example, March) and the horizontal row updates immediately. In Excel 365 it spills automatically; in older versions confirm with Ctrl+Shift+Enter.

Screenshot:



G	H	I	J	K	L	M	N	O	P	Q	R	S
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales	12000	15500	9800	22000	18400	14100	27500	19500	13650	15100	14000	17650
By Transpose Formula:												
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales	12000	15500	9800	22000	18400	14100	27500	19500	13650	15100	14000	17650

1.3 Using COUNTBLANK, COUNT & COUNTA

Task 1: Create a dataset with product names, sales numbers, and some blank cells.

💥 **Answer:** Product / Units Sold table: Chair 15; Table (blank); (blank product) 20; Storage "N/A"; Desk 10. It includes a blank number cell, a blank name cell and a text entry.

Task 2: Use the COUNT function to count the numerical values in the sales column.

💥 **Answer:** `=COUNT(...)` gives **3**. It counts only the numbers 15, 20 and 10.

Task 3: Apply the COUNTA function to count all non-blank cells in the product names column.

💥 **Answer:** `=COUNTA(...)` gives **4**, the non-empty product names (Chair, Table, Storage, Desk).

Task 4: Use the COUNTBLANK function to count the number of blank cells in the sales column.

💥 **Answer:** `=COUNTBLANK(...)` gives **1**, the empty Units Sold cell next to Table.

Task 5: Compare the results of COUNT, COUNTA, and COUNTBLANK functions.

💥 **Answer:** COUNT = 3 (numbers only). COUNTA = 4 (anything non-empty, including text such as "N/A"). COUNTBLANK = 1 (empty cells only). Together they show how complete the data is.

Screenshot:

Product	Units Sold
Chair	15
Table	
	20
Storage	N/A
Desk	10
COUNT:	3
COUNTA:	4
COUNTBLANK:	1

1.4 Quick Calculations using Status Bar

Task 1: Select a range of sales data and observe the sum, average, and count in the status bar.

📸 **Answer:** Selecting the 10 sales amounts shows Average 23,700, Count 22 and Sum 2,37,000 in the status bar. Count is higher because both the date and amount columns were selected.

Task 2: Compare the quick calculation results with the functions SUM, AVERAGE, and COUNT.

📸 **Answer:** `=SUM`, `=AVERAGE` and `=COUNT` on the same selection return the same values as the status bar, which confirms they match.

Task 3: Explore other calculations available in the status bar (e.g., Min, Max).

📸 **Answer:** Right-click the status bar and tick Minimum, Maximum, Numerical Count and others. They then appear next to Sum and Average.

Task 4: Use the status bar to quickly verify data analysis results.

📸 **Answer:** Select the monthly sales and the status bar shows Sum = 1,99,200 and Average = 16,600, which match the SUM and AVERAGE formula results.

Task 5: Explain the benefits of using the status bar for quick calculations.

📸 **Answer:** It gives instant results without writing formulas or using extra cells. It is useful for quick checks and validating formulas, and it never changes the sheet.

1.5 Calculating Growth Percentage YoY & Using POWER Formula

Task 1: Create a dataset of yearly revenue data for 5 years.

📸 **Answer:** 2021: 1,00,000; 2022: 1,20,000; 2023: 1,15,000; 2024: 1,45,000; 2025: 1,80,000.

Task 2: Calculate the year-over-year growth percentage for each year.

📸 **Answer:** `=(C3-C2)/C2` copied down. 2022: **20.00%**; 2023: **-4.17%**; 2024: **26.09%**; 2025: **24.14%**.

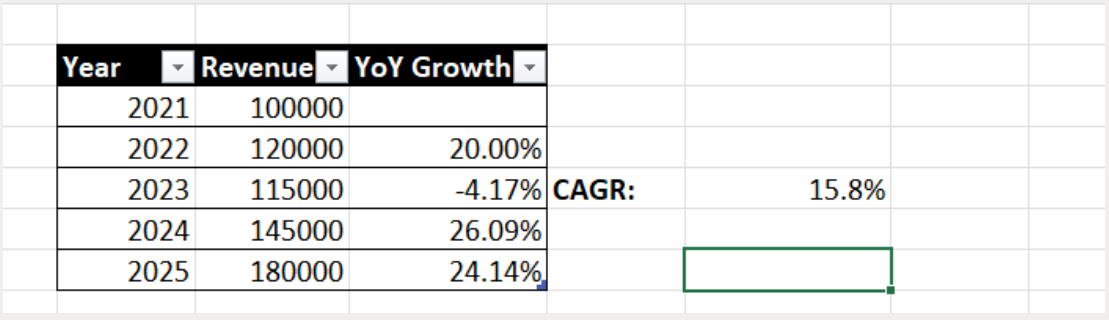
Screenshot:

Year	Revenue	YoY Growth
2021	100000	
2022	120000	20.00%
2023	115000	-4.17%
2024	145000	26.09%
2025	180000	24.14%

Task 3: Use the POWER formula to calculate the compound annual growth rate (CAGR).

📸 **Answer:** `=POWER(180000/100000,1/4)-1` gives **15.83%**. There are 4 growth periods between 2021 and 2025, so the exponent is 1/4.

Screenshot:

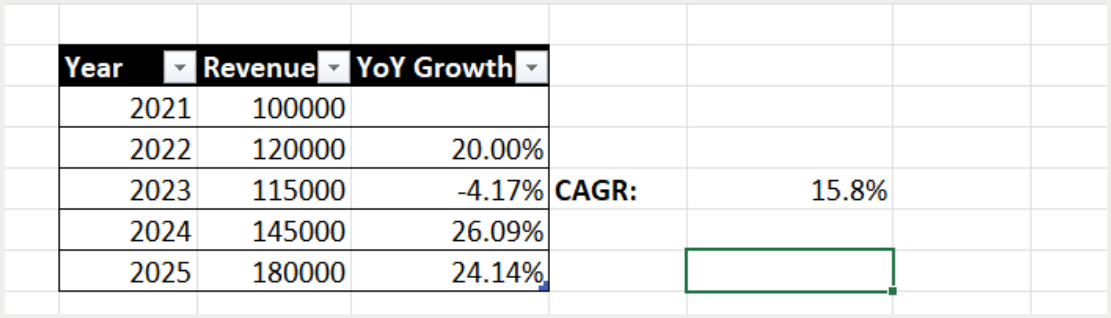


Year	Revenue	YoY Growth		
2021	100000			
2022	120000	20.00%		
2023	115000	-4.17%	CAGR:	15.8%
2024	145000	26.09%		
2025	180000	24.14%		

Task 4: Compare the manual growth percentage calculation with the POWER formula results.

📸 **Answer:** YoY shows the change between each pair of years, and it swings from -4.17% to 26.09%. CAGR gives one smoothed annual rate (15.83%) across the whole 5 years.

Screenshot:



Year	Revenue	YoY Growth		
2021	100000			
2022	120000	20.00%		
2023	115000	-4.17%	CAGR:	15.8%
2024	145000	26.09%		
2025	180000	24.14%		

Task 5: Explain the significance of growth percentage and CAGR in business analysis.

📸 **Answer:** YoY growth reveals short-term performance and problem years (like the 2023 dip). CAGR shows long-term growth, removes volatility and lets you compare businesses or periods fairly. It is used in forecasting and investor reporting.

Section 2: Formatting Options in Excel

2.1 Ways to Format Cells

Task 1: Format the sales data cells to have a currency format.

📸 **Answer:** Sales is formatted as ₹ currency with two decimals (for example ₹12,000.00), using Home → Number → Currency.

Task 2: Apply a conditional format to highlight sales figures greater than a specified amount.

📸 **Answer:** Home → Conditional Formatting → Highlight Cells Rules → Greater Than. The orange highlight marks the top months: Apr ₹22,000, Jul ₹27,500 and Aug ₹19,500.

Task 3: Use different font styles and sizes for the header row.

📸 **Answer:** Header cells use bold text with a larger or different font style than the body (Home → Font group).

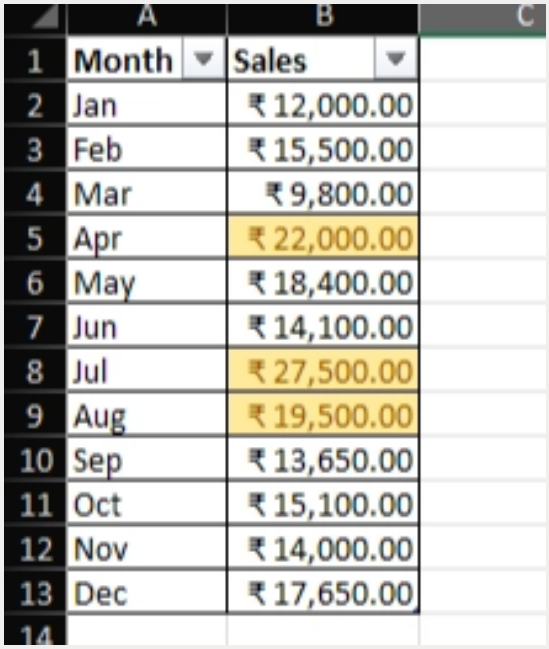
Task 4: Add borders to the table to improve readability.

📸 **Answer:** Home → Borders → All Borders is applied to the table, so rows and columns are clearly separated.

Task 5: Change the background color of the cells based on the sales performance.

📸 **Answer:** The high-performing months are filled orange (Apr, Jul, Aug), so the top months stand out from the rest.

Screenshot:



	A	B	C
1	Month	Sales	
2	Jan	₹ 12,000.00	
3	Feb	₹ 15,500.00	
4	Mar	₹ 9,800.00	
5	Apr	₹ 22,000.00	
6	May	₹ 18,400.00	
7	Jun	₹ 14,100.00	
8	Jul	₹ 27,500.00	
9	Aug	₹ 19,500.00	
10	Sep	₹ 13,650.00	
11	Oct	₹ 15,100.00	
12	Nov	₹ 14,000.00	
13	Dec	₹ 17,650.00	
14			

2.2 Formatting Numbers in Different Ways

Task 1: Format numbers to display as percentages.

💥 **Answer:** The YoY Growth column uses Percentage format with 2 decimals (20.00%, -4.17%, 26.09%, 24.14%).

Task 2: Use custom number formatting to display sales figures in thousands.

💥 **Answer:** Format Cells → Custom → `#,##0,"K"`. For example, 12,000 shows as 12K and 27,500 as 28K.

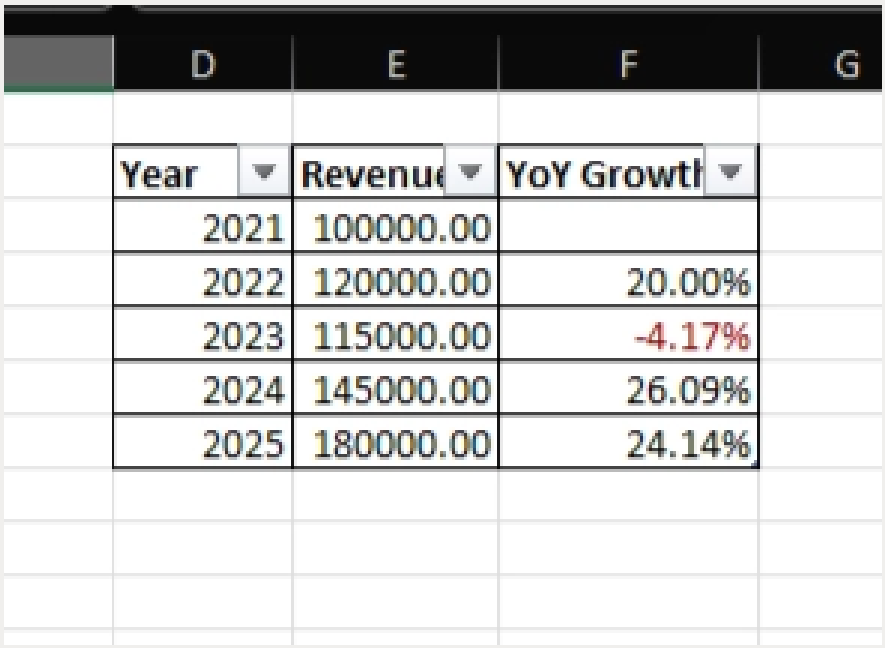
Task 3: Format negative numbers to appear in red.

💥 **Answer:** The -4.17% growth shows in red. Custom format: `0.00%;[Red]-0.00%`.

Task 4: Apply number formatting to show a fixed number of decimal places.

📸 **Answer:** Revenue is shown with exactly 2 decimals (100000.00, 120000.00 and so on) using the Increase or Decrease Decimal buttons.

Screenshot:



Year	Revenue	YoY Growth
2021	100000.00	
2022	120000.00	20.00%
2023	115000.00	-4.17%
2024	145000.00	26.09%
2025	180000.00	24.14%

Task 5: Experiment with scientific number formatting.

📸 **Answer:** Format Cells → Scientific with 2 decimals. For example, 199200 displays as 1.99E+05.

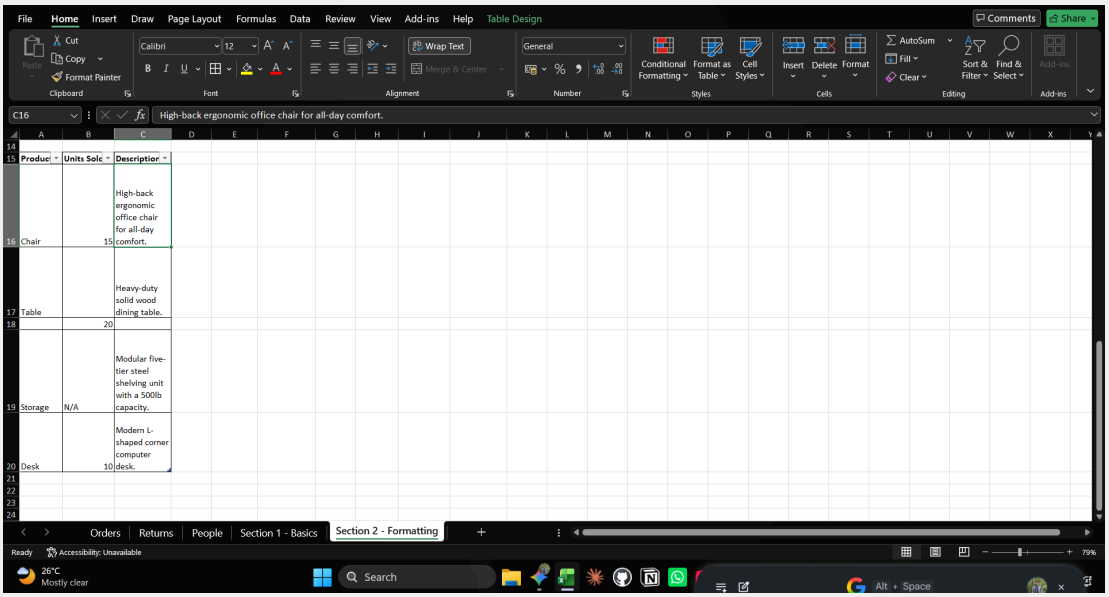
2.3 Formatting Lengthy Text using Wrap Text and Alignment

Task 1: Enter lengthy product descriptions and use the Wrap Text feature.

📸 **Answer:** Descriptions such as "High-back ergonomic office chair for all-day comfort." are entered in a Description column. Wrap Text is switched

on so the text wraps within the cell.

Screenshot:



Task 2: Align text to the center of the cells.

📸 **Answer:** Home → Center alignment is applied, so the wrapped descriptions are centred in their cells.

Task 3: Use the Merge & Center option for the table headers.

📸 **Answer:** Select the header cells above the table and click Home → Merge & Center. A single centred title spans the table.

Task 4: Adjust the row height and column width for better text display.

📸 **Answer:** Row height and column width are adjusted (drag the borders or use Format → AutoFit), so the wrapped descriptions are fully visible.

Task 5: Apply text rotation to the headers for a better presentation.

📸 **Answer:** Home → Orientation → Angle Counterclockwise (or Format Cells → Alignment → 45 degrees) is applied to the headers.

2.4 All About Format Painter

Task 1: Format a cell with a specific style (font, color, border).

📸 **Answer:** One cell is styled with bold text, a light-blue fill, centre alignment and a border.

Task 2: Use Format Painter to apply the same formatting to multiple cells.

📸 **Answer:** Select the styled cell, click Format Painter, then drag over the target cells. Double-click the brush to paint several areas in a row.

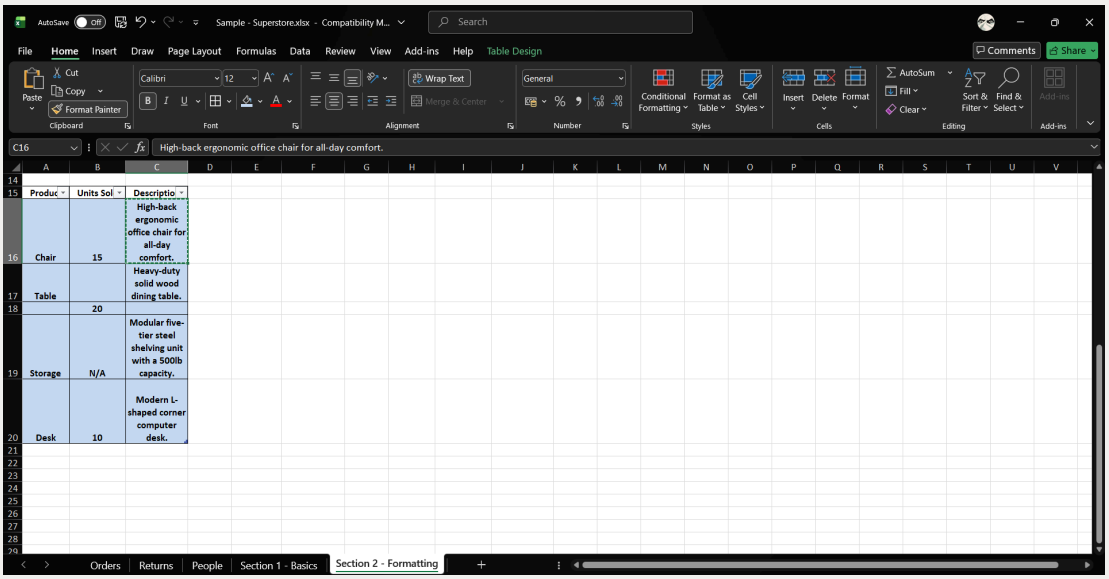
Task 3: Experiment with copying formatting from one worksheet to another.

📸 **Answer:** Click Format Painter on the source cell, switch to another sheet tab and click or drag the target cells. The formatting transfers across sheets.

Task 4: Apply Format Painter to a range of cells.

📸 **Answer:** The light-blue, bold, centred style is applied to the whole Product / Units Sold / Description table range in one drag.

Screenshot:



Task 5: Explain the advantages of using Format Painter in Excel.

📸 **Answer:** It copies fonts, fills, borders and number formats in one click. That saves time, keeps formatting consistent and avoids repeating many manual steps.

Section 3: Sort and Filter Option

3.1 Sorting Numbers and Dates in Different Ways

Task 1: Create a dataset of sales dates and amounts.

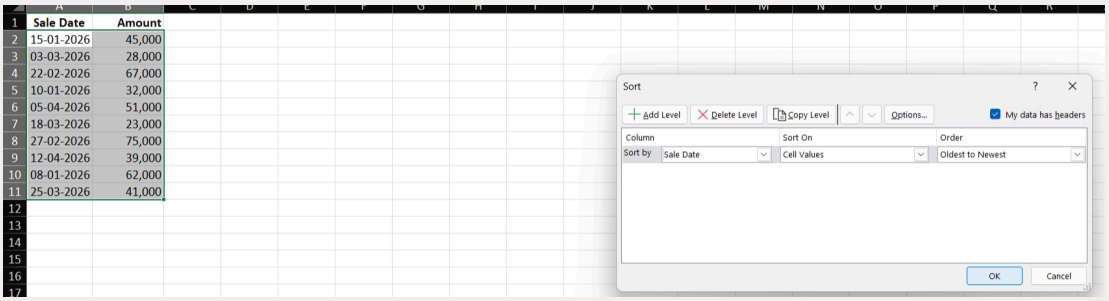
📸 **Answer:** 10 records of Sale Date and Amount, for example 15-01-2026 → 45,000 and 03-03-2026 → 28,000. Dates are formatted dd-mm-yyyy.

Screenshot:

Task 2: Sort the sales data by date in ascending order.

Answer: Data → Sort → Sale Date, Cell Values, Oldest to Newest. The result starts at 08-01-2026 and ends at 12-04-2026.

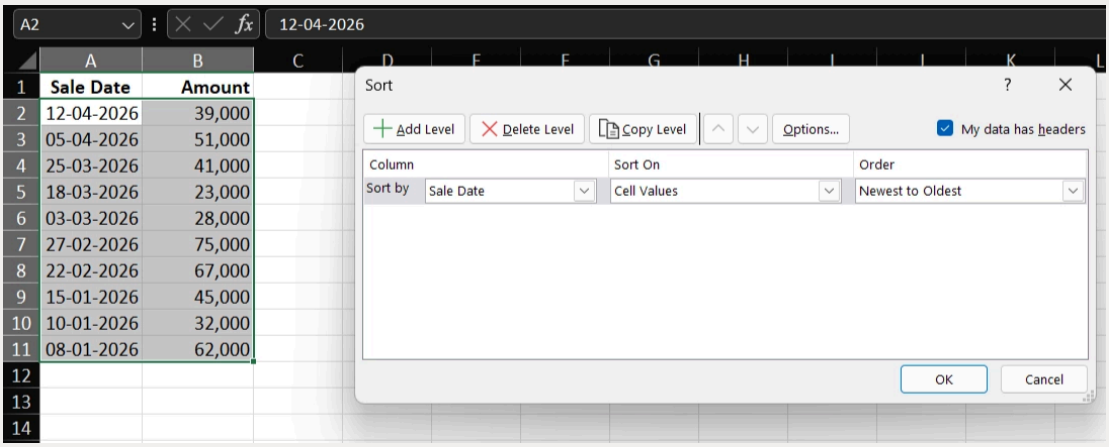
Screenshot:



Task 3: Sort the sales data by date in descending order.

Answer: Sort by Sale Date, Newest to Oldest. The list now runs from 12-04-2026 down to 08-01-2026.

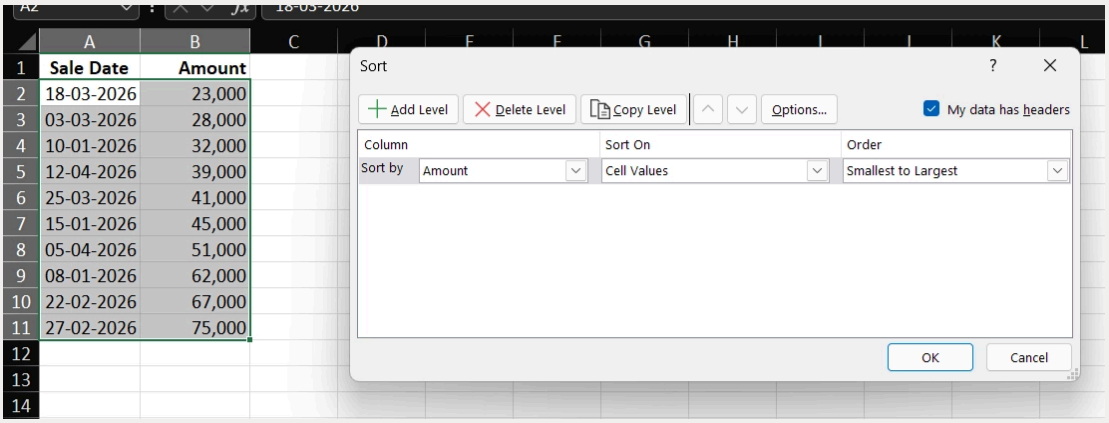
Screenshot:



Task 4: Sort the sales amounts in ascending order.

Answer: Sort by Amount, Smallest to Largest: 23,000 up to 75,000.

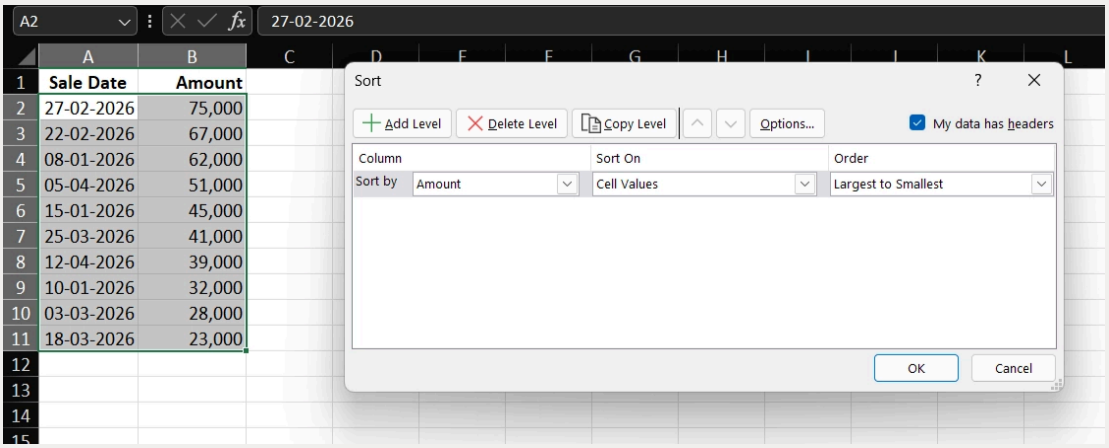
Screenshot:



Task 5: Sort the sales amounts in descending order.

Answer: Sort by Amount, Largest to Smallest: 75,000 down to 23,000.

Screenshot:



3.2 Sorting Text Alphabetically (A-Z) or (Z-A)

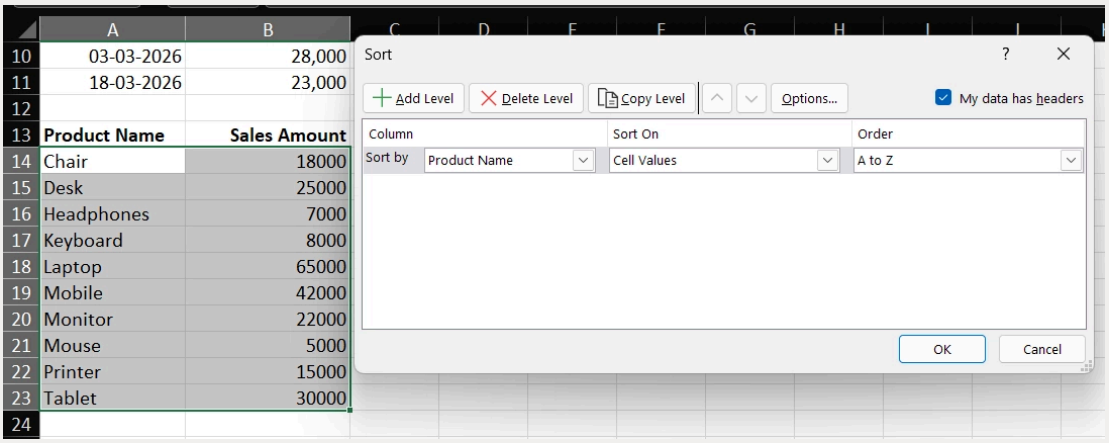
Task 1: Create a list of product names.

📸 **Answer:** Product Name and Sales Amount list: Chair, Desk, Headphones, Keyboard, Laptop, Mobile, Monitor, Mouse, Printer, Tablet.

Task 2: Sort the product names alphabetically from A to Z.

📸 **Answer:** Data → Sort → Product Name, A to Z. The list begins with Chair and ends with Tablet.

Screenshot:

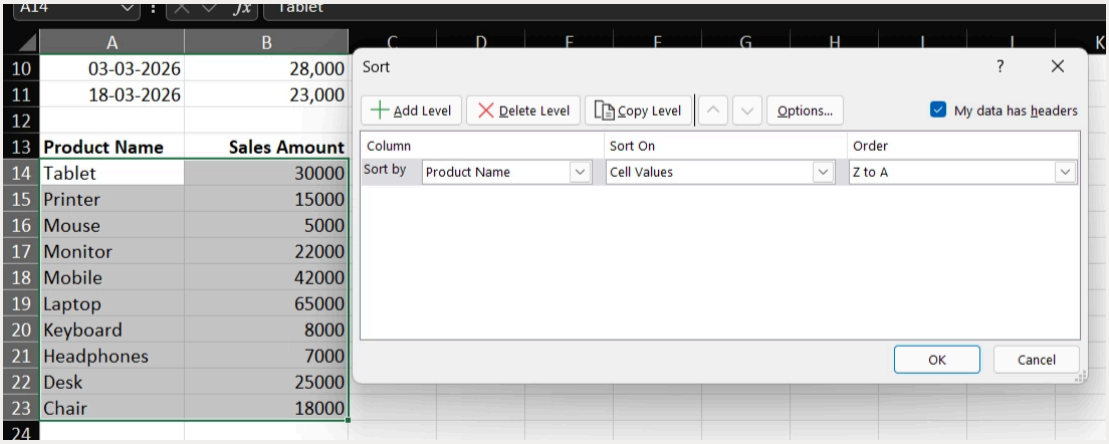


	A	B
10	03-03-2026	28,000
11	18-03-2026	23,000
12		
13	**Product Name**	**Sales Amount**
14	Chair	18000
15	Desk	25000
16	Headphones	7000
17	Keyboard	8000
18	Laptop	65000
19	Mobile	42000
20	Monitor	22000
21	Mouse	5000
22	Printer	15000
23	Tablet	30000
24		

Task 3: Sort the product names alphabetically from Z to A.

📸 **Answer:** Sort by Product Name, Z to A. The list begins with Tablet and ends with Chair.

Screenshot:



	A	B
10	03-03-2026	28,000
11	18-03-2026	23,000
12		
13	**Product Name**	**Sales Amount**
14	Tablet	30000
15	Printer	15000
16	Mouse	5000
17	Monitor	22000
18	Mobile	42000
19	Laptop	65000
20	Keyboard	8000
21	Headphones	7000
22	Desk	25000
23	Chair	18000
24		

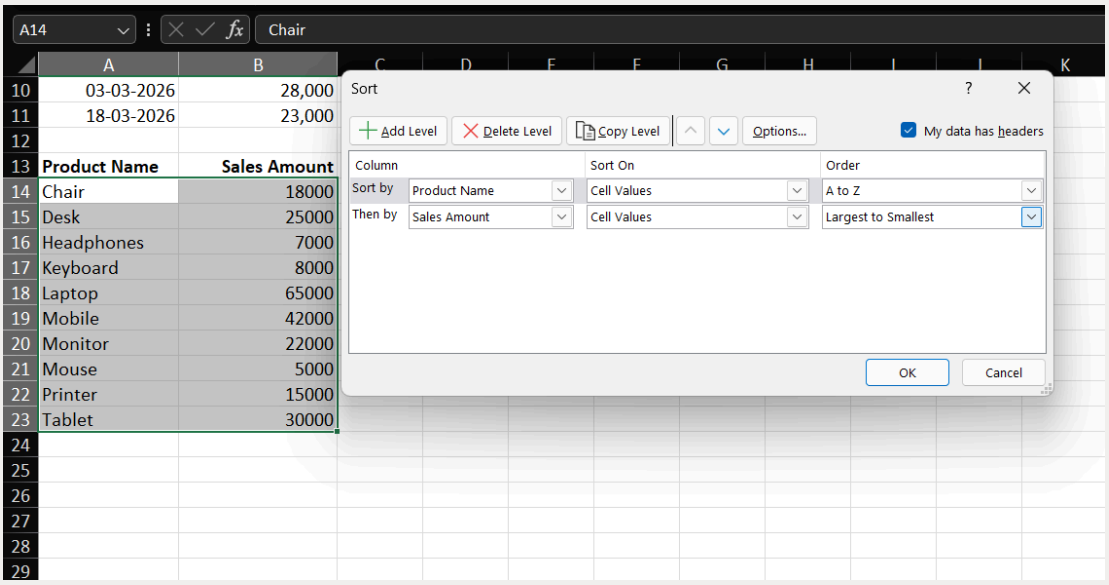
Task 4: Explain the impact of sorting on data analysis.

Answer: Sorting organises data so patterns, top and bottom values and duplicates are easy to spot. It also makes lookups and reviews faster.

Task 5: Combine sorting by product name and sales amount.

Answer: Two-level sort: Sort by Product Name (A to Z), then by Sales Amount (Largest to Smallest). The second level only matters when names repeat.

Screenshot:



	A	B
10	03-03-2026	28,000
11	18-03-2026	23,000
12		
13	**Product Name**	**Sales Amount**
14	Chair	18000
15	Desk	25000
16	Headphones	7000
17	Keyboard	8000
18	Laptop	65000
19	Mobile	42000
20	Monitor	22000
21	Mouse	5000
22	Printer	15000
23	Tablet	30000

3.3 Sorting by Cell Colour

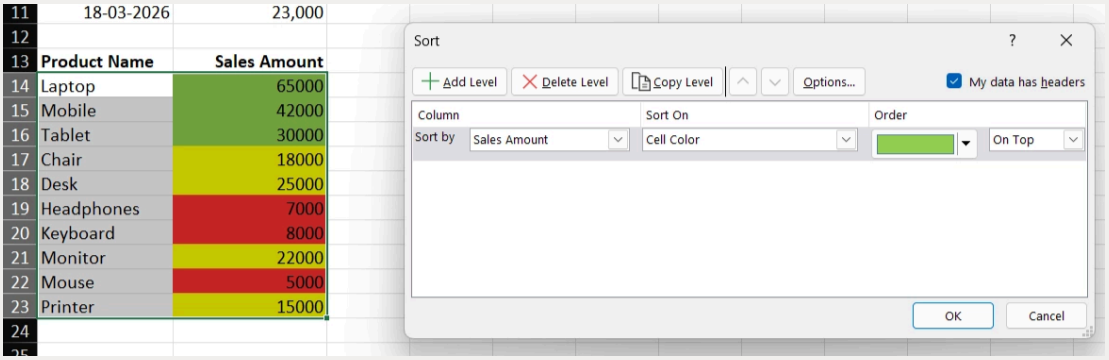
Task 1: Apply different background colors to cells based on sales performance.

Answer: Sales Amount cells are coloured green (30,000 and above), yellow (15,000 to 29,999) and red (below 15,000).

Task 2: Sort the data by cell color.

Answer: Data → Sort → Sales Amount, Sort On Cell Color, green On Top. Add more levels for yellow, then red to order all three.

Screenshot:



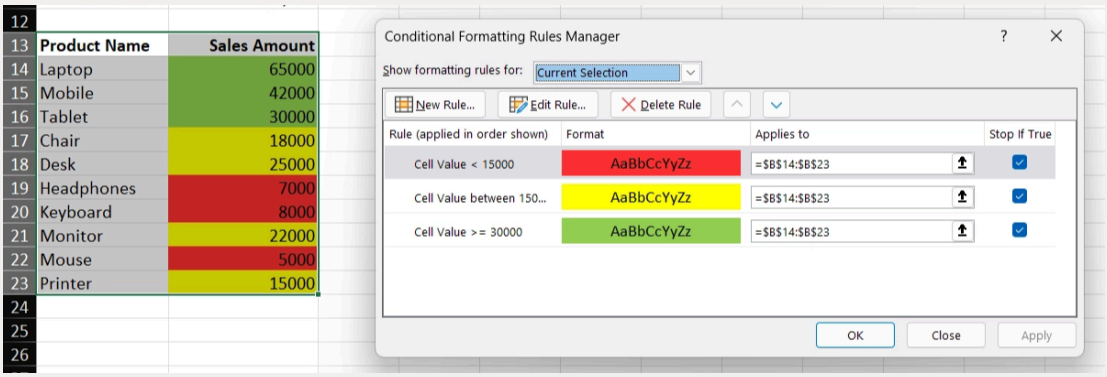
Task 3: Analyze the sorted data to identify trends.

Answer: Green (top performers): Laptop 65,000, Mobile 42,000, Tablet 30,000. Yellow (mid): Chair, Desk, Monitor, Printer. Red (weak): Headphones, Keyboard, Mouse. Computers and phones lead, and accessories lag.

Task 4: Use conditional formatting to apply colors and then sort.

Answer: Rules Manager shows three rules on the Sales Amount range: Cell Value less than 15000 is red, between 15000 and 29999 is yellow, and 30000 or more is green. The coloured cells were then sorted by cell colour.

Screenshot:



Task 5: Explain how sorting by color can enhance data visualization.

Answer: It groups items by performance tier, so the best and worst stand out together. It also makes colour-coded sheets easier to read and present.

3.4 2 Level Sorting & Tricks for Sorting

Task 1: Create a dataset with multiple columns (e.g., product name, sales, date).

Answer: Product Name, Sales Amount and Sale Date columns: Laptop 65,000 (12-Feb-26), Laptop 45,000 (three dates), Desk, Chair, Mobile, Monitor and so on.

Task 2: Perform a primary sort by sales and a secondary sort by date.

Answer: Sort by Sales Amount (Largest to Smallest), Then by Sale Date (Oldest to Newest). Ties are broken by date, for example the 45,000 rows run 05-Jan, 15-Jan, 25-Jan.

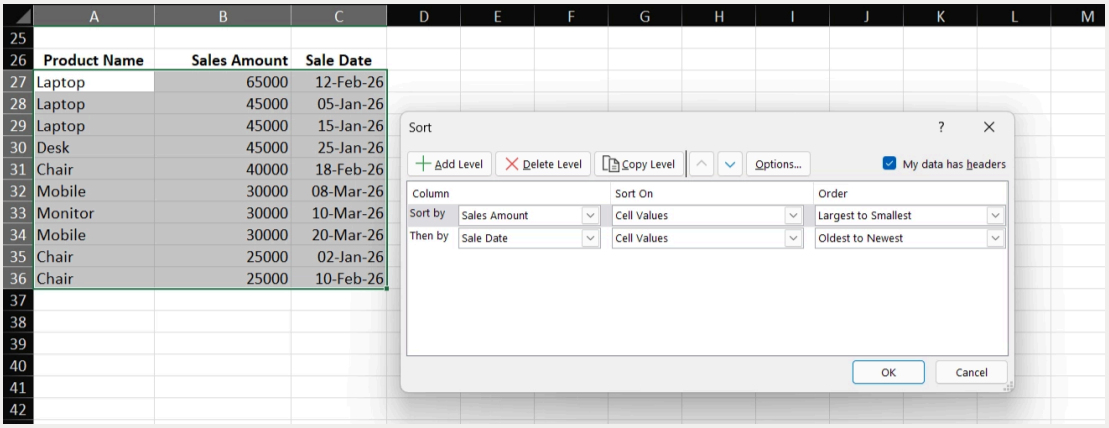
Task 3: Use custom sort options for more complex sorting criteria.

📸 **Answer:** In the Sort dialog use Add Level and Options (Case sensitive, left to right), or Order → Custom List (for example High, Medium, Low) to define your own sequence.

Task 4: Apply a multi-level sort to organize the data hierarchically.

📸 **Answer:** Sales Amount is the first level and Sale Date the second. The data is organised in a hierarchy: biggest sales first, then by oldest date within equal amounts.

Screenshot:



Task 5: Explain the importance of multi-level sorting in data analysis.

📸 **Answer:** It breaks ties and gives a predictable, repeatable order. Records can be ranked by one measure and refined by another, which is useful for reports.

Section 4: Working with Dates

Example values are used below. Replace them with the ones in your sheet.

4.1 Dates in Excel and Calculating the Duration Using Two Dates with Timestamp

Task 1: Enter start and end dates with timestamps for various events.

> 📸 **Answer:** Example: Start 01-01-2026 09:00 and End 15-03-2026 17:30, with several more events in the rows below. Format the cells as dd-mm-yyyy hh:mm.
>
> Screenshot:

Task 2: Calculate the duration between two dates using subtraction.

> 📸 **Answer:** `=End-Start` gives 73.35 days for the example (73 days plus 8.5 hours). Format as `[h]:mm` to show 1760:30 hours.
>
> Screenshot:

Task 3: Use the DATEDIF function to find the difference in days, months, and years.

> 📸 **Answer:** `=DATEDIF(Start,End,"d")` = 73 days; `"m"` = 2 months; `"y"` = 0 years. Use `"ym"` and `"md"` for the remaining months and days.
>
> Screenshot:

Task 4: Apply custom date formatting to display dates and times.

📸 **Answer:** Format Cells → Custom: `dd-mmm-yyyy hh:mm AM/PM` shows 01-Jan-2026 09:00 AM; `dddd, dd mmmm yyyy` shows Thursday, 01 January 2026.

Screenshot:

Task 5: Compare the manual duration calculation with the DATEDIF function results.

📸 **Answer:** Subtraction returns total days including the time part (73.35). DATEDIF returns whole units only (73 days, 2 months, 0 years) and ignores the time. DATEDIF is better for ages and tenures. Subtraction is better when the time of day matters.

Screenshot:

4.2 Using Date Formulas

Task 1: Use the TODAY function to display the current date.

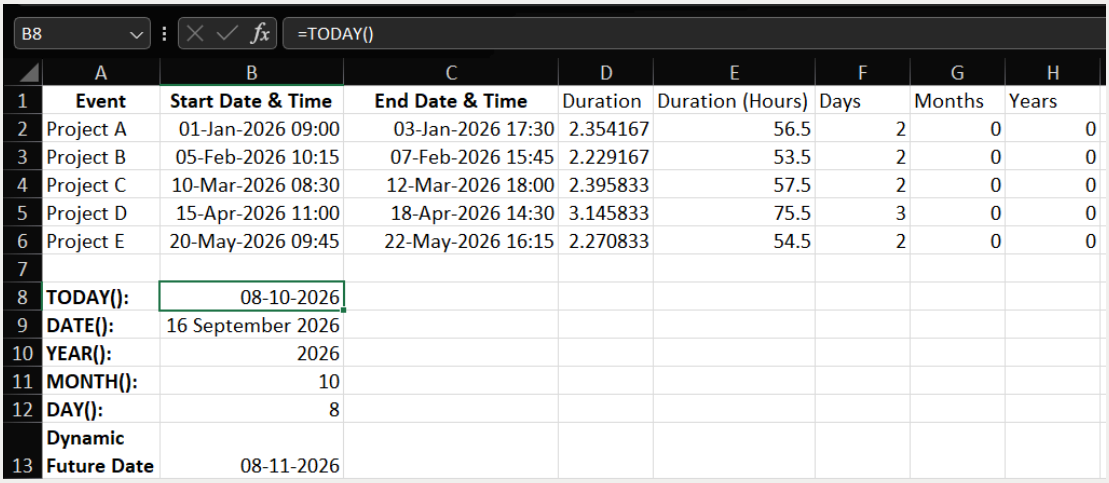
📸 **Answer:** `=TODAY()` shows the current date and updates every time the workbook opens or recalculates.

Task 2: Apply the DATE function to create a specific date.

📸 **Answer:** `=DATE(2026,9,18)` returns 18-09-2026.

Task 3: Use the YEAR, MONTH, and DAY functions to extract components from a date.

Answer: For 15-03-2026: `=YEAR(A2)` = 2026; `=MONTH(A2)` = 3; `=DAY(A2)` = 15.

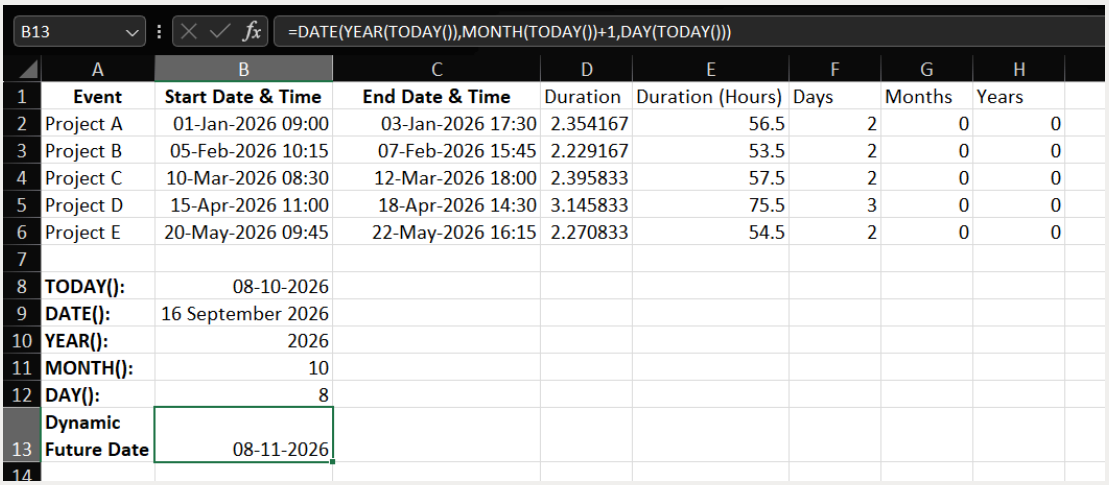


	A	B	C	D	E	F	G	H
1	**Event**	**Start Date & Time**	**End Date & Time**	Duration	Duration (Hours)	Days	Months	Years
2	Project A	01-Jan-2026 09:00	03-Jan-2026 17:30	2.354167	56.5	2	0	0
3	Project B	05-Feb-2026 10:15	07-Feb-2026 15:45	2.229167	53.5	2	0	0
4	Project C	10-Mar-2026 08:30	12-Mar-2026 18:00	2.395833	57.5	2	0	0
5	Project D	15-Apr-2026 11:00	18-Apr-2026 14:30	3.145833	75.5	3	0	0
6	Project E	20-May-2026 09:45	22-May-2026 16:15	2.270833	54.5	2	0	0
7								
8	**TODAY():**	08-10-2026						
9	**DATE():**	16 September 2026						
10	**YEAR():**	2026						
11	**MONTH():**	10						
12	**DAY():**	8						
13	**Dynamic Future Date**	08-11-2026						

Task 4: Combine date functions to create dynamic date calculations.

Answer: `=DATE(YEAR(TODAY()),MONTH(TODAY())+1,DAY(TODAY()))` gives the first day of next month. `=EOMONTH(TODAY(),0)` gives the month-end. `=TODAY()-A2` gives the days elapsed since a date.

Screenshot:



	A	B	C	D	E	F	G	H
1	**Event**	**Start Date & Time**	**End Date & Time**	Duration	Duration (Hours)	Days	Months	Years
2	Project A	01-Jan-2026 09:00	03-Jan-2026 17:30	2.354167	56.5	2	0	0
3	Project B	05-Feb-2026 10:15	07-Feb-2026 15:45	2.229167	53.5	2	0	0
4	Project C	10-Mar-2026 08:30	12-Mar-2026 18:00	2.395833	57.5	2	0	0
5	Project D	15-Apr-2026 11:00	18-Apr-2026 14:30	3.145833	75.5	3	0	0
6	Project E	20-May-2026 09:45	22-May-2026 16:15	2.270833	54.5	2	0	0
7								
8	**TODAY():**	08-10-2026						
9	**DATE():**	16 September 2026						
10	**YEAR():**	2026						
11	**MONTH():**	10						
12	**DAY():**	8						
13	**Dynamic Future Date**	08-11-2026						
14								

Task 5: Explain the significance of using date functions in data analysis.

📸 **Answer:** They automate time-based analysis such as ageing, deadlines, monthly and yearly grouping and trend reports. They update automatically and avoid typing dates by hand.

Section 5: Logical Functions

5.1 Basic Logical Formulas

Task 1: Use the IF function to classify sales as "High" or "Low" based on a threshold.

📸 **Answer:** Sales Category column: `=IF(B2>500,"High","Low")`. For example 731.94 gives High and 14.62 gives Low.

D2 =IF(B2>=500,"High","Low")

	A	B	C	D
1	**Product Name**	**Sales**	**Profit**	**Sales Category**
2	Bush Somerset Collection Bookcase	261.96	41.9136	Low
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582	High
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714	Low
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031	High
6	Eldon Fold 'N Roll Cart System	22.368	2.5164	Low
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694	Low
8	Newell 322	7.28	1.9656	Low
9	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	High
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825	Low
11	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47	Low
12	Chromcraft Rectangular Conference Tables	1706.184	85.3092	High
13	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	High
14	Xerox 1967	15.552	5.4432	Low
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922	Low
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858	Low
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816	Low
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176	High
19	Fellowes Super Stor/Drawer	55.5	9.99	Low
20	Newell 341	8.56	2.4824	Low

Task 2: Apply the AND function to check if sales meet multiple criteria.

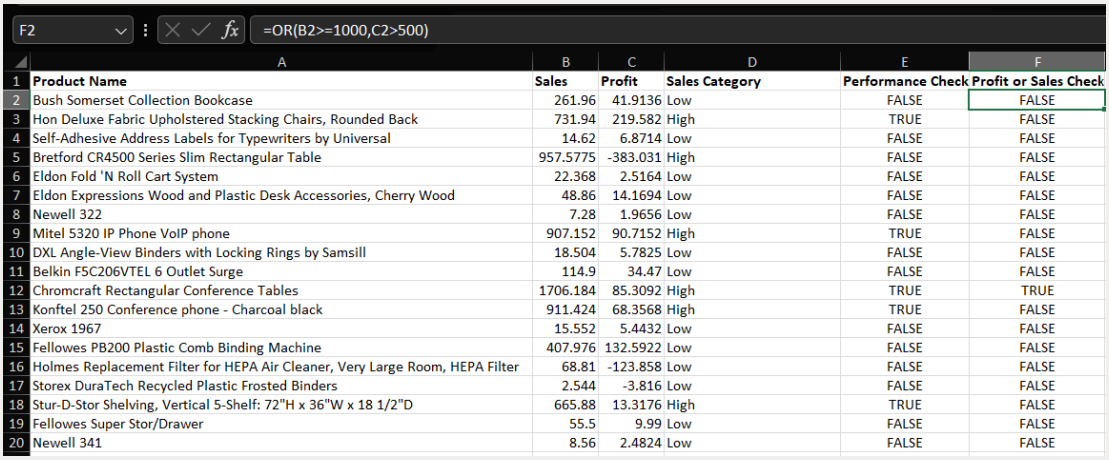
Answer: Performance Check column: `=AND(B2>500,C2>0)` is TRUE only when sales exceed 500 and profit is positive. For example, 957.58 sales with -383 profit gives FALSE.

E2 =AND(B2>=500,C2>0)

	A	B	C	D	E
1	**Product Name**	**Sales**	**Profit**	**Sales Category**	**Performance Check**
2	Bush Somerset Collection Bookcase	261.96	41.9136	Low	FALSE
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582	High	TRUE
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714	Low	FALSE
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031	High	FALSE
6	Eldon Fold 'N Roll Cart System	22.368	2.5164	Low	FALSE
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694	Low	FALSE
8	Newell 322	7.28	1.9656	Low	FALSE
9	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	High	TRUE
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825	Low	FALSE
11	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47	Low	FALSE
12	Chromcraft Rectangular Conference Tables	1706.184	85.3092	High	TRUE
13	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	High	TRUE
14	Xerox 1967	15.552	5.4432	Low	FALSE
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922	Low	FALSE
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858	Low	FALSE
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816	Low	FALSE
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176	High	TRUE
19	Fellowes Super Stor/Drawer	55.5	9.99	Low	FALSE
20	Newell 341	8.56	2.4824	Low	FALSE

Task 3: Use the OR function to determine if any of the conditions are met.

Answer: Profit Check column: `=OR(B2>=1000,C2>500)` is TRUE if either condition holds. Only the 1,706.18 sale returns TRUE.

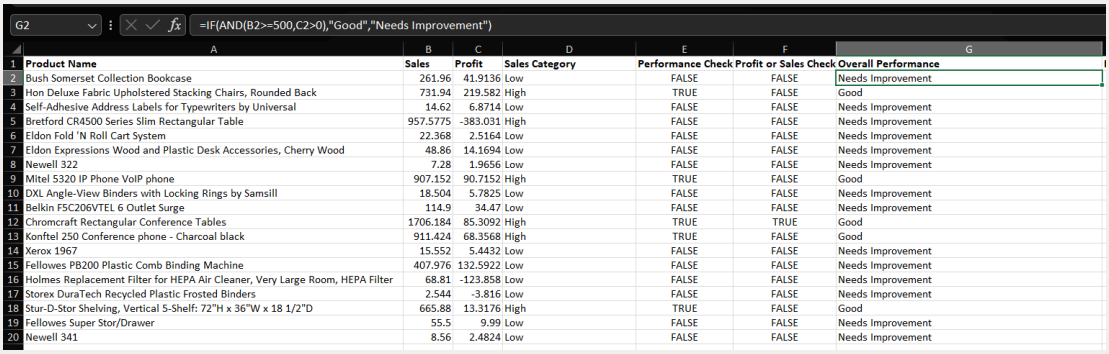


F2 =OR(B2>=1000,C2>500)

	A	B	C	D	E	F
1	**Product Name**	**Sales**	**Profit**	**Sales Category**	**Performance Check**	**Profit or Sales Check**
2	Bush Somerset Collection Bookcase	261.96	41.9136	Low	FALSE	FALSE
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582	High	TRUE	FALSE
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714	Low	FALSE	FALSE
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031	High	FALSE	FALSE
6	Eldon Fold 'N Roll Cart System	22.368	2.5164	Low	FALSE	FALSE
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694	Low	FALSE	FALSE
8	Newell 322	7.28	1.9656	Low	FALSE	FALSE
9	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	High	TRUE	FALSE
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825	Low	FALSE	FALSE
11	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47	Low	FALSE	FALSE
12	Chromcraft Rectangular Conference Tables	1706.184	85.3092	High	TRUE	TRUE
13	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	High	TRUE	FALSE
14	Xerox 1967	15.552	5.4432	Low	FALSE	FALSE
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922	Low	FALSE	FALSE
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858	Low	FALSE	FALSE
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816	Low	FALSE	FALSE
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176	High	TRUE	FALSE
19	Fellowes Super Stor/Drawer	55.5	9.99	Low	FALSE	FALSE
20	Newell 341	8.56	2.4824	Low	FALSE	FALSE

Task 4: Combine IF and AND functions for complex logical tests.

Answer: `=IF(AND(B2>500,C2>0),"Strong","Review")` labels a row Strong only if sales and profit both pass.

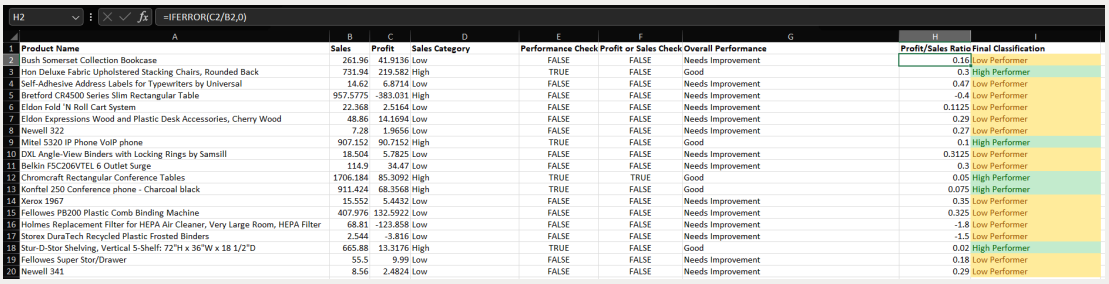


G2 =IF(AND(B2>=500,C2>0),"Good","Needs Improvement")

	A	B	C	D	E	F	G
1	**Product Name**	**Sales**	**Profit**	**Sales Category**	**Performance Check**	**Profit or Sales Check**	**Overall Performance**
2	Bush Somerset Collection Bookcase	261.96	41.9136	Low	FALSE	FALSE	Needs Improvement
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582	High	TRUE	FALSE	Good
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714	Low	FALSE	FALSE	Needs Improvement
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031	High	FALSE	FALSE	Needs Improvement
6	Eldon Fold 'N Roll Cart System	22.368	2.5164	Low	FALSE	FALSE	Needs Improvement
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694	Low	FALSE	FALSE	Needs Improvement
8	Newell 322	7.28	1.9656	Low	FALSE	FALSE	Needs Improvement
9	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	High	TRUE	FALSE	Good
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825	Low	FALSE	FALSE	Needs Improvement
11	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47	Low	FALSE	FALSE	Needs Improvement
12	Chromcraft Rectangular Conference Tables	1706.184	85.3092	High	TRUE	TRUE	Good
13	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	High	TRUE	FALSE	Good
14	Xerox 1967	15.552	5.4432	Low	FALSE	FALSE	Needs Improvement
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922	Low	FALSE	FALSE	Needs Improvement
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858	Low	FALSE	FALSE	Needs Improvement
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816	Low	FALSE	FALSE	Needs Improvement
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176	High	TRUE	FALSE	Good
19	Fellowes Super Stor/Drawer	55.5	9.99	Low	FALSE	FALSE	Needs Improvement
20	Newell 341	8.56	2.4824	Low	FALSE	FALSE	Needs Improvement

Task 5: Use the IFERROR function to handle errors in formulas.

Screenshot:



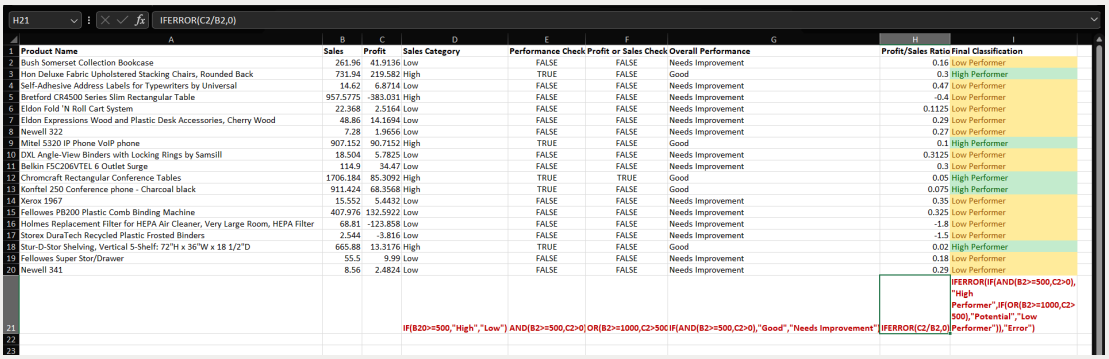
	A	B	C	D	E	F	G	H	I
1	Product Name	Sales	Profit	Sales Category	Performance Check	Profit or Sales Check	Overall Performance	Profit/Sales Ratio	Final Classification
2	Bush Somerset Collection Bookcase	261.96	41.9136	Low	FALSE	FALSE	Needs Improvement	0.16	Low Performer
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582	High	TRUE	FALSE	Good	0.3	High Performer
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714	Low	FALSE	FALSE	Needs Improvement	0.47	Low Performer
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031	High	FALSE	FALSE	Needs Improvement	-0.4	Low Performer
6	Eldon Fold 'N Roll Cart System	22.368	2.5164	Low	FALSE	FALSE	Needs Improvement	0.1125	Low Performer
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694	Low	FALSE	FALSE	Needs Improvement	0.29	Low Performer
8	Newell 322	7.28	1.9656	Low	FALSE	FALSE	Needs Improvement	0.27	Low Performer
9	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	High	TRUE	FALSE	Good	0.1	High Performer
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825	Low	FALSE	FALSE	Needs Improvement	0.3125	Low Performer
11	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47	Low	FALSE	FALSE	Needs Improvement	0.3	Low Performer
12	Chromcraft Rectangular Conference Tables	1706.184	85.3092	High	TRUE	TRUE	Good	0.05	High Performer
13	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	High	TRUE	FALSE	Good	0.075	High Performer
14	Xerox 1967	15.552	5.4432	Low	FALSE	FALSE	Needs Improvement	0.35	Low Performer
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922	Low	FALSE	FALSE	Needs Improvement	0.325	Low Performer
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858	Low	FALSE	FALSE	Needs Improvement	-1.8	Low Performer
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816	Low	FALSE	FALSE	Needs Improvement	-1.5	Low Performer
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176	High	TRUE	FALSE	Good	0.02	High Performer
19	Fellowes Super Stor/Drawer	55.5	9.99	Low	FALSE	FALSE	Needs Improvement	0.18	Low Performer
20	Newell 341	8.56	2.4824	Low	FALSE	FALSE	Needs Improvement	0.29	Low Performer

5.2 Using Multiple Logical Formulas - IF, AND, OR & Implementing on Text, Numbers, & Errors Including IFERROR

Task 4: Implement logical tests on numerical data.

Answer: The tests above run on the numeric Sales and Profit columns, for example `=IF(C2<0,"Low","High")`.

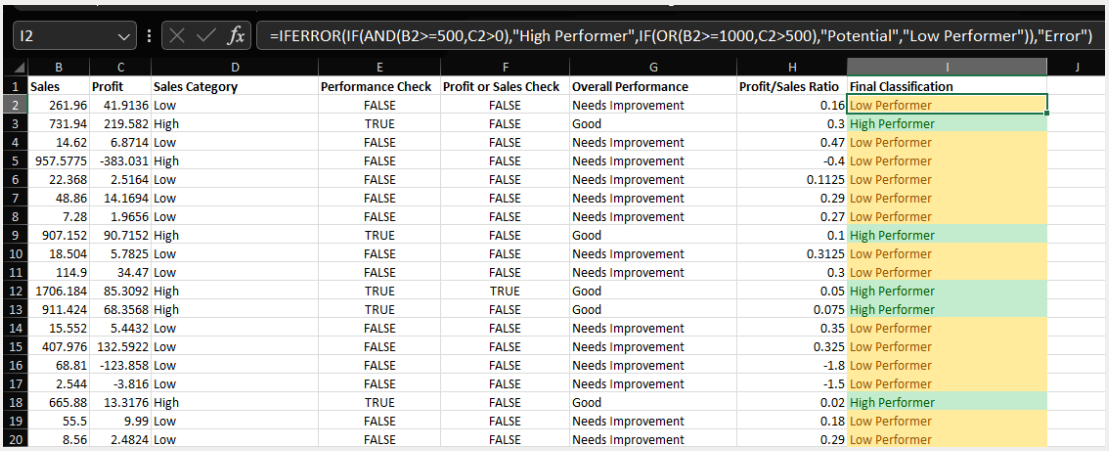
Screenshot:



	A	B	C	D	E	F	G	H	I
1	Product Name	Sales	Profit	Sales Category	Performance Check	Profit or Sales Check	Overall Performance	Profit/Sales Ratio	Final Classification
2	Bush Somerset Collection Bookcase	261.96	41.9136	Low	FALSE	FALSE	Needs Improvement	0.16	Low Performer
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582	High	TRUE	FALSE	Good	0.3	High Performer
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714	Low	FALSE	FALSE	Needs Improvement	0.47	Low Performer
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031	High	FALSE	FALSE	Needs Improvement	-0.4	Low Performer
6	Eldon Fold 'N Roll Cart System	22.368	2.5164	Low	FALSE	FALSE	Needs Improvement	0.1125	Low Performer
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694	Low	FALSE	FALSE	Needs Improvement	0.29	Low Performer
8	Newell 322	7.28	1.9656	Low	FALSE	FALSE	Needs Improvement	0.27	Low Performer
9	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	High	TRUE	FALSE	Good	0.1	High Performer
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825	Low	FALSE	FALSE	Needs Improvement	0.3125	Low Performer
11	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47	Low	FALSE	FALSE	Needs Improvement	0.3	Low Performer
12	Chromcraft Rectangular Conference Tables	1706.184	85.3092	High	TRUE	TRUE	Good	0.05	High Performer
13	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	High	TRUE	FALSE	Good	0.075	High Performer
14	Xerox 1967	15.552	5.4432	Low	FALSE	FALSE	Needs Improvement	0.35	Low Performer
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922	Low	FALSE	FALSE	Needs Improvement	0.325	Low Performer
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858	Low	FALSE	FALSE	Needs Improvement	-1.8	Low Performer
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816	Low	FALSE	FALSE	Needs Improvement	-1.5	Low Performer
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176	High	TRUE	FALSE	Good	0.02	High Performer
19	Fellowes Super Stor/Drawer	55.5	9.99	Low	FALSE	FALSE	Needs Improvement	0.18	Low Performer
20	Newell 341	8.56	2.4824	Low	FALSE	FALSE	Needs Improvement	0.29	Low Performer
21				IF(B20>=500,"High","Low")	AND(B2>=500,C2>0)	OR(B2>=1000,C2>500	IF(AND(B2>=500,C2>0),"Good","Needs Improvement"	IFERROR(C2/B2,0)	IFERROR(IF(AND(B2>=500,C2>0),"High Performer",IF(OR(B2>=1000,C2>500),"Potential","Low Performer")),"Error")
22									
23									

Task 5: Combine IF, AND, OR, and IFERROR to create a comprehensive logical test for your dataset.

Screenshot:



I2 =IFERROR(IF(AND(B2>=500,C2>0),"High Performer",IF(OR(B2>=1000,C2>500),"Potential","Low Performer")),"Error")

	B	C	D	E	F	G	H	I	J
1	Sales	Profit	Sales Category	Performance Check	Profit or Sales Check	Overall Performance	Profit/Sales Ratio	Final Classification	
2	261.96	41.9136	Low	FALSE	FALSE	Needs Improvement	0.16	Low Performer	
3	731.94	219.582	High	TRUE	FALSE	Good	0.3	High Performer	
4	14.62	6.8714	Low	FALSE	FALSE	Needs Improvement	0.47	Low Performer	
5	957.5775	-383.031	High	FALSE	FALSE	Needs Improvement	-0.4	Low Performer	
6	22.368	2.5164	Low	FALSE	FALSE	Needs Improvement	0.1125	Low Performer	
7	48.86	14.1694	Low	FALSE	FALSE	Needs Improvement	0.29	Low Performer	
8	7.28	1.9656	Low	FALSE	FALSE	Needs Improvement	0.27	Low Performer	
9	907.152	90.7152	High	TRUE	FALSE	Good	0.1	High Performer	
10	18.504	5.7825	Low	FALSE	FALSE	Needs Improvement	0.3125	Low Performer	
11	114.9	34.47	Low	FALSE	FALSE	Needs Improvement	0.3	Low Performer	
12	1706.184	85.3092	High	TRUE	TRUE	Good	0.05	High Performer	
13	911.424	68.3568	High	TRUE	FALSE	Good	0.075	High Performer	
14	15.552	5.4432	Low	FALSE	FALSE	Needs Improvement	0.35	Low Performer	
15	407.976	132.5922	Low	FALSE	FALSE	Needs Improvement	0.325	Low Performer	
16	68.81	-123.858	Low	FALSE	FALSE	Needs Improvement	-1.8	Low Performer	
17	2.544	-3.816	Low	FALSE	FALSE	Needs Improvement	-1.5	Low Performer	
18	665.88	13.3176	High	TRUE	FALSE	Good	0.02	High Performer	
19	55.5	9.99	Low	FALSE	FALSE	Needs Improvement	0.18	Low Performer	
20	8.56	2.4824	Low	FALSE	FALSE	Needs Improvement	0.29	Low Performer	

Section 6: Practical Data Cleaning

6.1 Removing Duplicate Values and Error Values

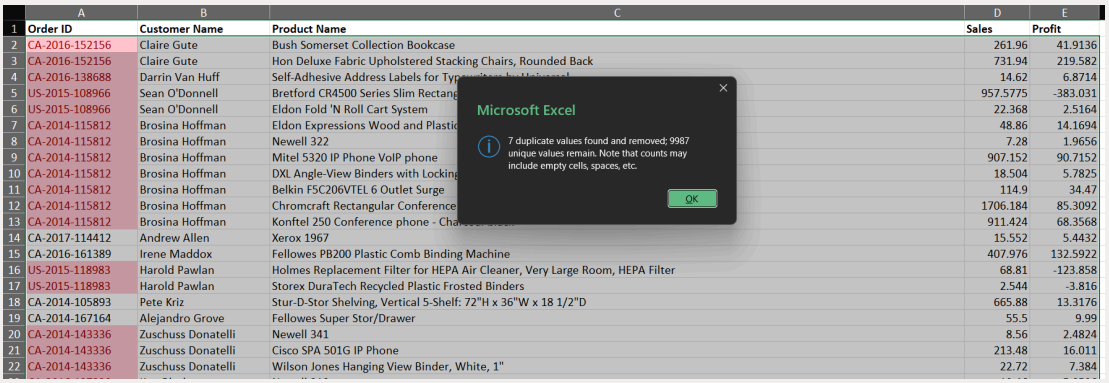
Task 1: Create a dataset with duplicate values.

📸 **Answer:** The Orders data has repeated Order IDs (for example CA-2016-152156 and CA-2014-115812 each appear multiple times).

Task 2: Use the Remove Duplicates feature to clean up the data.

📸 **Answer:** Data → Remove Duplicates. Excel reports "7 duplicate values found and removed; 9987 unique values remain".

Screenshot:



	A	B	C	D	E
1	Order ID	Customer Name	Product Name	Sales	Profit
2	CA-2016-152156	Claire Gute	Bush Somerset Collection Bookcase	261.96	41.9136
3	CA-2016-152156	Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582
4	CA-2016-138688	Darrin Van Huff	Self-Adhesive Address Labels for Typ	14.62	6.8714
5	US-2015-108966	Sean O'Donnell	Bretford CR4500 Series Slim Rectang	957.5775	-383.031
6	US-2015-108966	Sean O'Donnell	Eldon Fold 'N Roll Cart System	22.368	2.5164
7	CA-2014-115812	Brosina Hoffman	Eldon Expressions Wood and Plastic	48.86	14.1694
8	CA-2014-115812	Brosina Hoffman	Newell 322	7.28	1.9656
9	CA-2014-115812	Brosina Hoffman	Mitel 5320 IP Phone VoIP phone	907.152	90.7152
10	CA-2014-115812	Brosina Hoffman	DXL Angle-View Binders with Locking	18.504	5.7825
11	CA-2014-115812	Brosina Hoffman	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47
12	CA-2014-115812	Brosina Hoffman	Chromcraft Rectangular Conference	1706.184	85.3092
13	CA-2014-115812	Brosina Hoffman	Konftel 250 Conference phone - Cha	911.424	68.3568
14	CA-2017-114412	Andrew Allen	Xerox 1967	15.552	5.4432
15	CA-2016-161389	Irene Maddox	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922
16	US-2015-118983	Harold Pawlan	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858
17	US-2015-118983	Harold Pawlan	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816
18	CA-2014-105893	Pete Kriz	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176
19	CA-2014-167164	Alejandro Grove	Fellowes Super Stor/Drawer	55.5	9.99
20	CA-2014-143336	Zuschuss Donatelli	Newell 341	8.56	2.4824
21	CA-2014-143336	Zuschuss Donatelli	Cisco SPA 501G IP Phone	213.48	16.011
22	CA-2014-143336	Zuschuss Donatelli	Wilson Jones Hanging View Binder, White, 1"	22.72	7.384

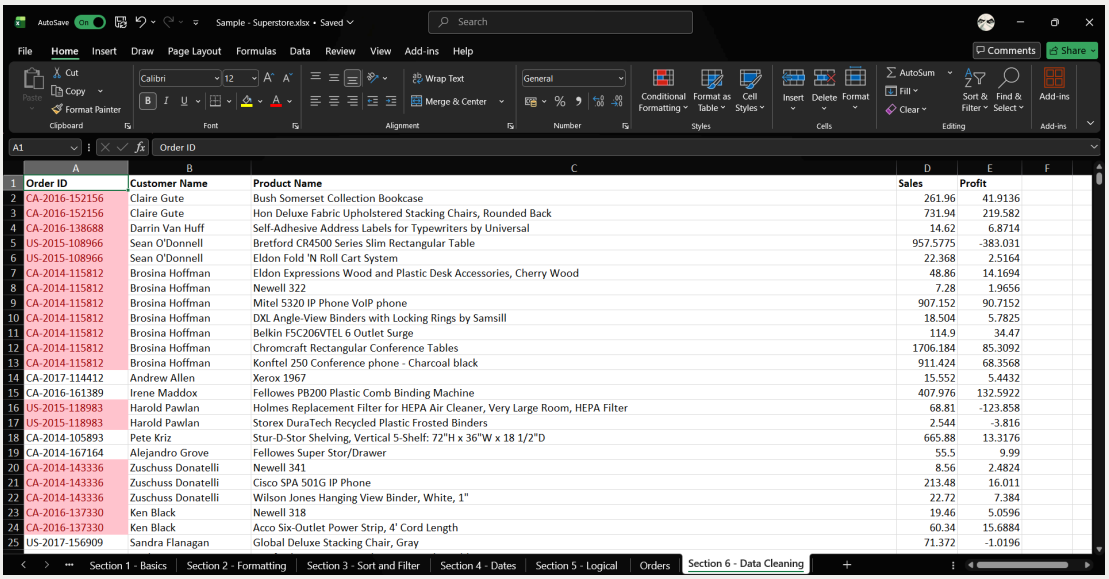
Task 3: Identify and remove error values from the dataset.

📸 **Answer:** Home → Find & Select → Go To Special → Formulas → Errors selects every error cell. Then delete them or wrap the source formula in IFERROR.

Task 4: Use conditional formatting to highlight duplicate values.

📸 **Answer:** Home → Conditional Formatting → Highlight Cells Rules → Duplicate Values. Repeated Order IDs are shaded pink.

Screenshot:

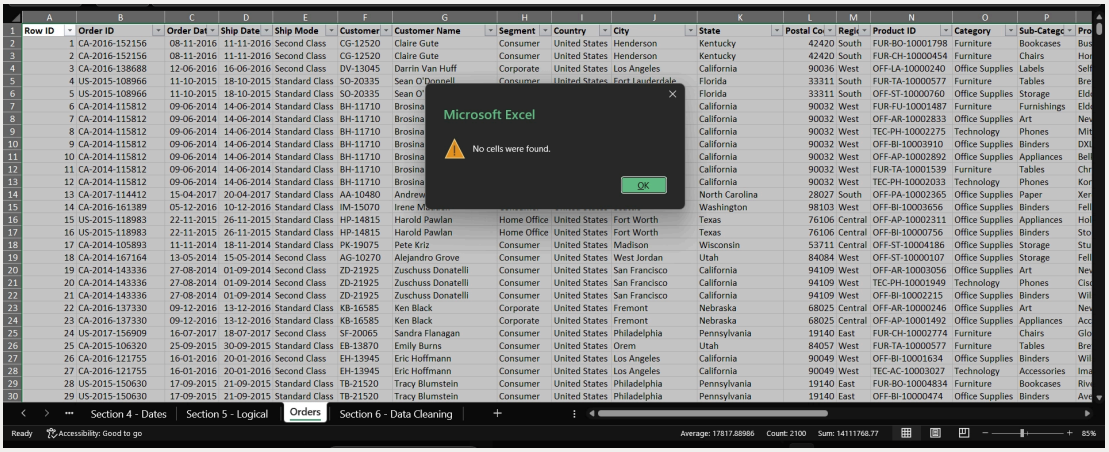


Order ID	Customer Name	Product Name	Sales	Profit
CA-2016-152156	Claire Gute	Bush Somerset Collection Bookcase	261.96	41.9136
CA-2016-152156	Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	219.582
CA-2016-138688	Darrin Van Huff	Self-Adhesive Address Labels for Typewriters by Universal	14.62	6.8714
US-2015-108966	Sean O'Donnell	Bretford CR4500 Series Slim Rectangular Table	957.5775	-383.031
US-2015-108966	Sean O'Donnell	Eldon Fold 'N Roll Cart System	22.368	2.5164
CA-2014-115812	Brosina Hoffman	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	14.1694
CA-2014-115812	Brosina Hoffman	Newell 322	7.28	1.9656
CA-2014-115812	Brosina Hoffman	Mitel 5320 IP Phone VoIP phone	907.152	90.7152
CA-2014-115812	Brosina Hoffman	DXL Angle-View Binders with Locking Rings by Samsill	18.504	5.7825
CA-2014-115812	Brosina Hoffman	Belkin F5C206VTEL 6 Outlet Surge	114.9	34.47
CA-2014-115812	Brosina Hoffman	Chromcraft Rectangular Conference Tables	1706.184	85.3092
CA-2014-115812	Brosina Hoffman	Konftel 250 Conference phone - Charcoal black	911.424	68.3568
CA-2017-114412	Andrew Allen	Xerox 1967	15.552	5.4432
CA-2016-161389	Irene Maddox	Fellowes PB200 Plastic Comb Binding Machine	407.976	132.5922
US-2015-118983	Harold Pawlan	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	-123.858
US-2015-118983	Harold Pawlan	Storex DuraTech Recycled Plastic Frosted Binders	2.544	-3.816
CA-2014-105893	Pete Kriz	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	13.3176
CA-2014-167164	Alejandro Grove	Fellowes Super Stor/Drawer	55.5	9.99
CA-2014-143336	Zuschuss Donatelli	Newell 341	8.56	2.4824
CA-2014-143336	Zuschuss Donatelli	Cisco SPA 501G IP Phone	213.48	16.011
CA-2014-143336	Zuschuss Donatelli	Wilson Jones Hanging View Binder, White, 1"	22.72	7.384
CA-2016-137330	Ken Black	Newell 318	19.46	5.0596
CA-2016-137330	Ken Black	Acco Six-Outlet Power Strip, 4' Cord Length	60.34	15.6884
US-2017-156909	Sandra Flanagan	Global Deluxe Stacking Chair, Gray	71.372	-1.0196

Task 5: Verify the data cleanliness after removing duplicates and errors.

📸 **Answer:** After cleaning, 9987 unique rows remain (the 7 duplicates were removed). The duplicate highlight can be re-applied to confirm no further repeats, and Go To Special → Errors should return "No cells were found".

Screenshot:



6.2 Using Data Cleaning Formulas UPPER, PROPER, LOWER, TRIM, VALUE, LEN, RIGHT, LEFT & MID

Task 1: Use the UPPER function to convert text to uppercase.

> 💥📷 **Answer:** `=UPPER(B2)` turns "Claire Gute" into **CLAIRE GUTE**.

Task 2: Apply the PROPER function to format text with proper case.

> 💥📷 **Answer:** `=PROPER(B2)` gives **Claire Gute**; "SEAN O'DONNELL" becomes "Sean O'Donnell".

Task 3: Use the LOWER function to convert text to lowercase.

> 💥📷 **Answer:** `=LOWER(B2)` gives **claire gute**.

Task 4: Apply the TRIM function to remove extra spaces from text.

> 💥📷 **Answer:** `=TRIM(B2)` returns the Clean Name column, with leading, trailing and doubled spaces removed.

Task 5: Use VALUE, LEN, RIGHT, LEFT, and MID functions for various data cleaning tasks.

> 💥📷 **Answer:** `=LEN(B2)` gives Name Length (11 for Claire Gute). `=LEFT(B2,5)` gives "Clair". `=RIGHT(B2,5)` gives " Gute". `=MID(B2,3,5)` gives "aire". `=VALUE(N2)` converts the text Postal Code into the number 42420.
>
> Screenshot:



D	E	F	G	H	I	J	K	L	M	N
Sales	Profit	UPPER Name	PROPER Name	LOWER Name	Clean Name	Name Length	First 5 Char	Last 5 Char	Middle Char	Postal Code
261.96	41.9136	CLAIRE GUTE	Claire Gute	claire gute	Claire Gute	11	Clair	Gute	aire	42420
731.94	219.582	CLAIRE GUTE	Claire Gute	claire gute	Claire Gute	11	Clair	Gute	aire	42420
14.62	6.8714	DARRIN VAN HUFF	Darrin Van Huff	darrin van huff	Darrin Van Huff	15	Darri	Huff	rrin	90036
957.5775	-383.031	SEAN O'DONNELL	Sean O'Donnell	sean o'donnell	Sean O'Donnell	14	Sean	nnell	an O'	33311
22.368	2.5164	SEAN O'DONNELL	Sean O'Donnell	sean o'donnell	Sean O'Donnell	14	Sean	nnell	an O'	33311
48.86	14.1694	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
7.28	1.9656	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
907.152	90.7152	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
18.504	5.7825	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
114.9	34.47	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
1706.184	85.3092	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
911.424	68.3568	BROSINA HOFFMAN	Brosina Hoffman	brosina hoffman	Brosina Hoffman	15	Brosi	ffman	osina	90032
15.552	5.4432	ANDREW ALLEN	Andrew Allen	andrew allen	Andrew Allen	12	Andre	Allen	drew	28027
407.976	132.5922	IRENE MADDOX	Irene Maddox	irene maddox	Irene Maddox	12	Irene	addox	ene M	98103
68.81	-123.858	HAROLD PAWLAN	Harold Pawlan	harold pawlan	Harold Pawlan	13	Harol	awlan	rold	76106
2.544	-3.816	HAROLD PAWLAN	Harold Pawlan	harold pawlan	Harold Pawlan	13	Harol	awlan	rold	76106
665.88	13.3176	PETE KRIZ	Pete Kriz	pete kriz	Pete Kriz	9	Pete	Kriz	te Kr	53711
55.5	9.99	ALEJANDRO GROVE	Alejandro Grove	alejandro grove	Alejandro Grove	15	Aleja	Grove	ejand	84084
8.56	2.4824	ZUSCHUSS DONATELLI	Zuschuss Donatelli	zuschuss donatelli	Zuschuss Donatelli	18	Zusch	telli	schus	94109
213.48	16.011	ZUSCHUSS DONATELLI	Zuschuss Donatelli	zuschuss donatelli	Zuschuss Donatelli	18	Zusch	telli	schus	94109
22.72	7.384	ZUSCHUSS DONATELLI	Zuschuss Donatelli	zuschuss donatelli	Zuschuss Donatelli	18	Zusch	telli	schus	94109
19.46	5.0596	KEN BLACK	Ken Black	ken black	Ken Black	9	Ken B	Black	n Bla	68025
60.34	15.6884	KEN BLACK	Ken Black	ken black	Ken Black	9	Ken B	Black	n Bla	68025
71.372	-1.0196	SANDRA FLANAGAN	Sandra Flanagan	sandra flanagan	Sandra Flanagan	15	Sandr	nagan	ndra	19140

6.3 SEARCH vs FIND and SEARCH & MID

Task 1: Use the SEARCH function to find a substring within a text string.



Answer: `=SEARCH("text",A2)` returns the position of the text, ignoring case. The SEARCH Position column shows values such as 5, 4, 14 and 9.

Task 2: Apply the FIND function to locate the position of a substring.



Answer: `=FIND("text",A2)` returns the position, but it is case-sensitive. The FIND Position column shows the same values.

Task 3: Compare the results of SEARCH and FIND functions.



Answer: Both columns return identical positions here because the searched text has the same letter case. If the case differed, FIND would return #VALUE! while SEARCH would still work.

Task 4: Combine SEARCH and MID functions to extract a substring.



Answer: Last name: `=MID(B2,SEARCH(" ",B2)+1,LEN(B2))`. SEARCH locates the space and MID returns everything after it. "Claire Gute" gives "Gute".

Task 5: Explain the differences and use cases for SEARCH vs FIND.



Answer: SEARCH is not case-sensitive and supports wildcards (? and *). FIND is case-sensitive and has no wildcards. Use SEARCH for flexible matching and FIND when exact case matters.

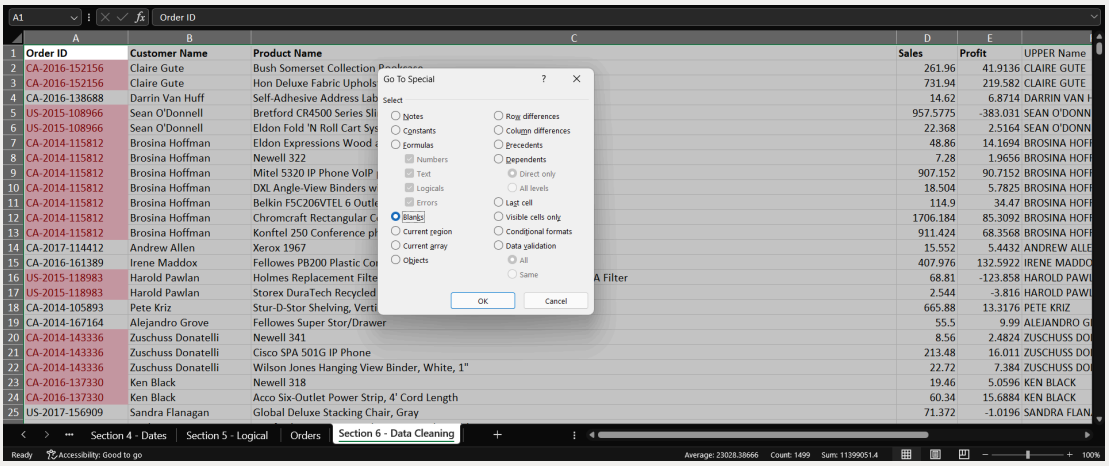
6.4 Using Go To (Special) Technique

Task 1: Use the Go To (Special) technique to select cells with specific attributes (e.g., blanks, constants).



Answer: Home → Find & Select → Go To Special opens the dialog with options such as Blanks, Constants, Formulas and Conditional formats. Blanks was selected on the data range.

Screenshot:



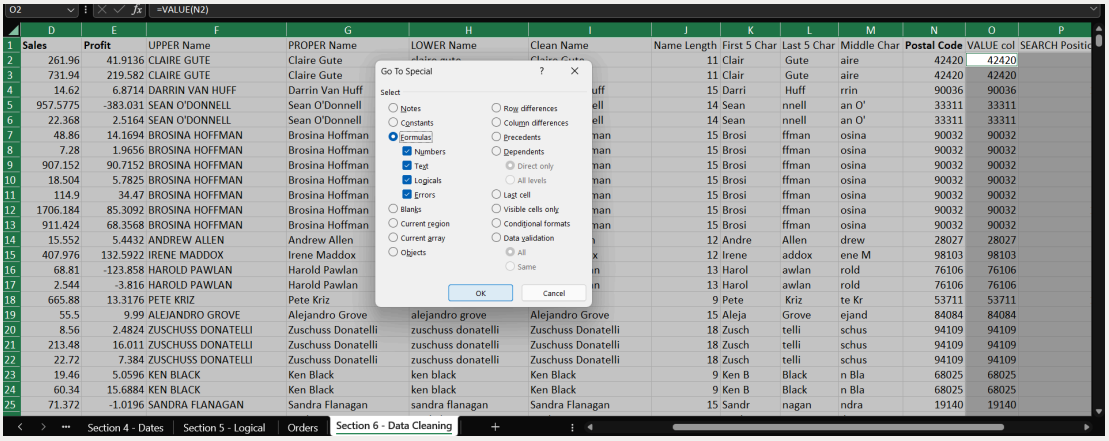
Task 2: Apply the technique to find and replace blank cells.

Answer: With Blanks selected, type a value (for example "N/A") and press Ctrl+Enter to fill every blank. On the Orders sheet Excel showed "No cells were found", so the data has no blanks.

Task 3: Use Go To (Special) to highlight cells with formulas.

Answer: Go To Special → Formulas (Numbers, Text, Logicals and Errors ticked) selects the formula columns UPPER, PROPER, LOWER, Clean Name, Name Length and the extraction columns, and leaves the constants untouched.

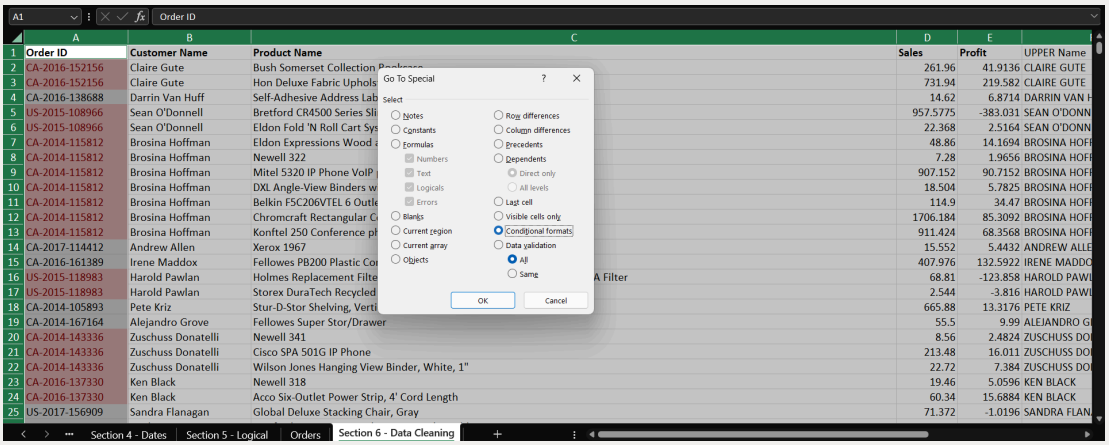
Screenshot:



Task 4: Select cells with conditional formatting using Go To (Special).

Answer: Go To Special → Conditional formats → All selects every cell with a conditional-format rule, here the pink duplicate Order IDs.

Screenshot:



Task 5: Explain the benefits of using Go To (Special) for data cleaning.

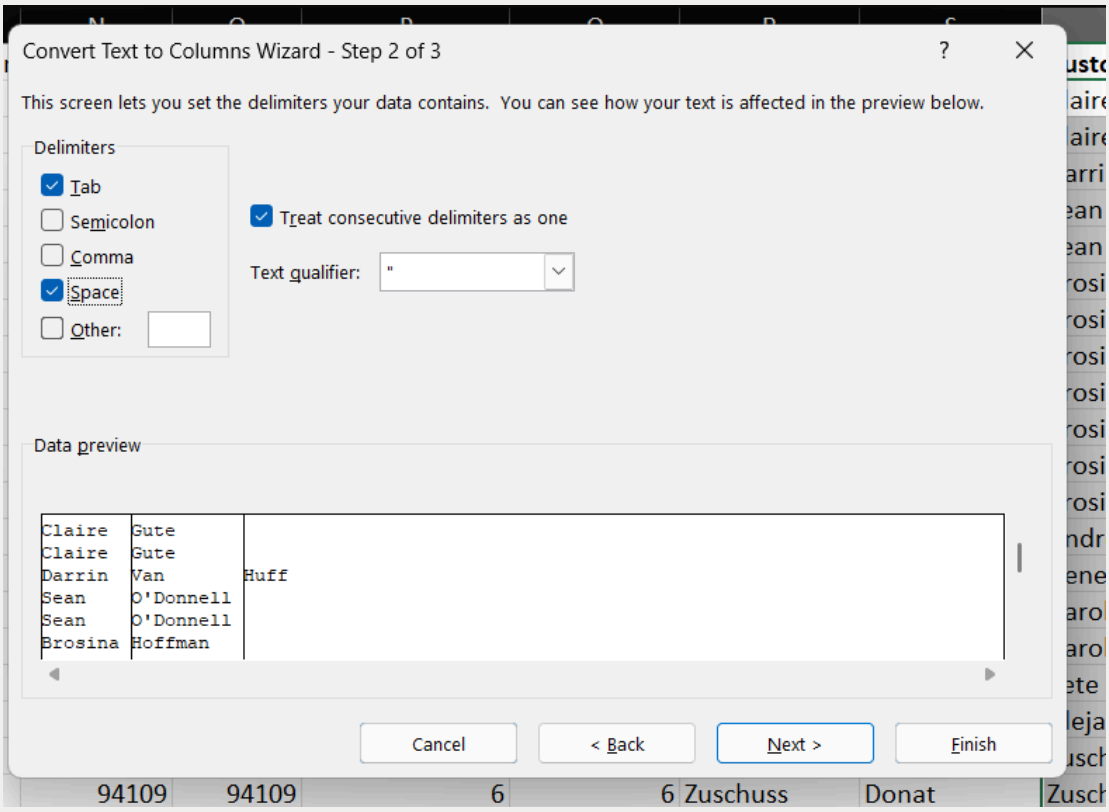
Answer: It selects blanks, errors, formulas, constants or formatted cells in one step, even in large sheets. That makes bulk fixes quick and avoids missing cells.

6.5 Splitting Data Values Using Text to Column Feature

Task 1: Create a dataset with concatenated text values (e.g., "FirstName LastName").

Answer: The Customer Name column holds combined values such as "Claire Gute" and "Sean O'Donnell". Order ID values like CA-2016-152156 are also combined text.

Screenshot:



Task 2: Use the Text to Columns feature to split the data into separate columns.

📸 **Answer:** Data → Text to Columns → Delimited → tick Space → Finish. "Claire Gute" splits into Claire and Gute.

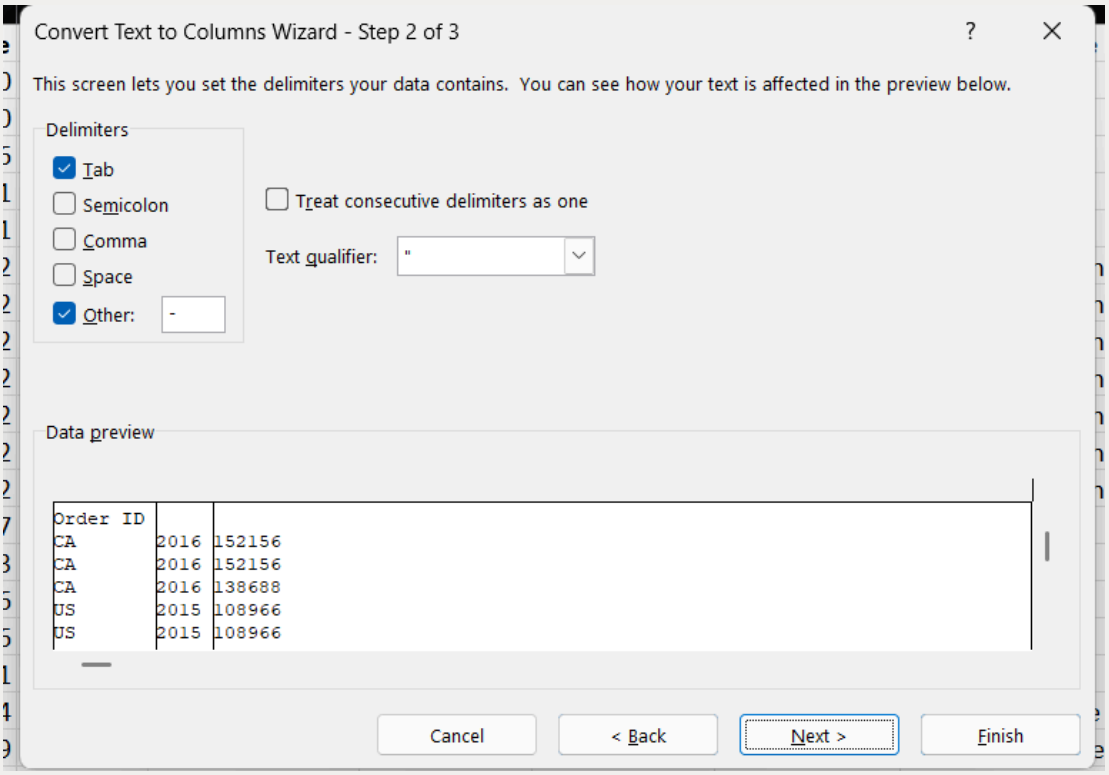
Screenshot:

Customer Name			
Claire Gute	Claire	Gute	
Claire Gute	Claire	Gute	
Darrin Van Huff	Darrin	Van	Huff
Sean O'Donnell	Sean	O'Donnell	
Sean O'Donnell	Sean	O'Donnell	
Brosina Hoffman	Brosina	Hoffman	
Brosina Hoffman	Brosina	Hoffman	
Brosina Hoffman	Brosina	Hoffman	
Brosina Hoffman	Brosina	Hoffman	
Brosina Hoffman	Brosina	Hoffman	
Brosina Hoffman	Brosina	Hoffman	
Brosina Hoffman	Brosina	Hoffman	
Andrew Allen	Andrew	Allen	
Irene Maddox	Irene	Maddox	
Harold Pawlan	Harold	Pawlan	
Harold Pawlan	Harold	Pawlan	
Pete Kriz	Pete	Kriz	
Alejandro Grove	Alejandro	Grove	
Zuschuss Donatelli	Zuschuss	Donatelli	
Zuschuss Donatelli	Zuschuss	Donatelli	
Zuschuss Donatelli	Zuschuss	Donatelli	
Ken Black	Ken	Black	
Ken Black	Ken	Black	
Sandra Flanagan	Sandra	Flanagan	

Task 3: Experiment with different delimiters (e.g., space, comma).

📸 **Answer:** A Space delimiter splits names. The Other delimiter "-" splits Order ID CA-2016-152156 into CA, 2016 and 152156. Comma and semicolon work the same way.

Screenshot:



Task 4: Apply Text to Columns to a larger dataset for practice.

Answer: The Order ID column from the Orders data (thousands of rows) was split into three columns in one operation.

Screenshot:

X	Y	Z	AA
Order ID	Order ID		
CA-2016-152156	CA	2016	152156
CA-2016-152156	CA	2016	152156
CA-2016-138688	CA	2016	138688
US-2015-108966	US	2015	108966
US-2015-108966	US	2015	108966
CA-2014-115812	CA	2014	115812
CA-2014-115812	CA	2014	115812
CA-2014-115812	CA	2014	115812
CA-2014-115812	CA	2014	115812
CA-2014-115812	CA	2014	115812
CA-2014-115812	CA	2014	115812
CA-2014-115812	CA	2014	115812
CA-2017-114412	CA	2017	114412
CA-2016-161389	CA	2016	161389
US-2015-118983	US	2015	118983
US-2015-118983	US	2015	118983
CA-2014-105893	CA	2014	105893
CA-2014-167164	CA	2014	167164
CA-2014-143336	CA	2014	143336
CA-2014-143336	CA	2014	143336
CA-2014-143336	CA	2014	143336
CA-2016-137330	CA	2016	137330
CA-2016-137330	CA	2016	137330
US-2017-156909	US	2017	156909

Task 5: Explain the advantages of using Text to Columns for data manipulation.

📸 **Answer:** It splits combined text in one step without formulas. It handles thousands of rows and makes filtering, sorting and grouping on each

part possible.

6.6 Using CONCATENATE and Ampersand (&) to Join Values

Task 1: Create a dataset with separate first and last names.

📸 **Answer:** First Name and Last Name columns (for example Claire / Gute, Sean / O'Donnell, Ken / Black).

Task 2: Use the CONCATENATE function to join first and last names.

📸 **Answer:** `=CONCATENATE(R2," ",S2)` gives "Claire Gute".

Task 3: Apply the ampersand (&) operator to achieve the same result.

📸 **Answer:** `=R2&" "&S2` gives "Claire Gute".

Screenshot:

R	S	T	U	V	W
First Name	**Last Name**	**Concatenated name**	**& Name**		
Claire	Gute	Claire Gute	Claire Gute		
Claire	Gute	Claire Gute	Claire Gute		**Concatenated name:**
Darrin	Van H	Darrin Van H	Darrin Van H		CONCATENATE(R2," ",S2)
Sean	O'Don	Sean O'Don	Sean O'Don		
Sean	O'Don	Sean O'Don	Sean O'Don		**& Name:**
Brosina	Hoffm	Brosina Hoffm	Brosina Hoffm		R2&" "&S2
Brosina	Hoffm	Brosina Hoffm	Brosina Hoffm		
Brosina	Hoffm	Brosina Hoffm	Brosina Hoffm		
Brosina	Hoffm	Brosina Hoffm	Brosina Hoffm		
Brosina	Hoffm	Brosina Hoffm	Brosina Hoffm		
Brosina	Hoffm	Brosina Hoffm	Brosina Hoffm		
Andrew	Allen	Andrew Allen	Andrew Allen		
Irene	Maddo	Irene Maddo	Irene Maddo		
Harold	Pawla	Harold Pawla	Harold Pawla		
Harold	Pawla	Harold Pawla	Harold Pawla		
Pete	Kriz	Pete Kriz	Pete Kriz		
Alejandro	Grove	Alejandro Grove	Alejandro Grove		
Zuschuss	Donat	Zuschuss Donat	Zuschuss Donat		
Zuschuss	Donat	Zuschuss Donat	Zuschuss Donat		
Zuschuss	Donat	Zuschuss Donat	Zuschuss Donat		
Ken	Black	Ken Black	Ken Black		
Ken	Black	Ken Black	Ken Black		
Sandra	Flana	Sandra Flana	Sandra Flana		

Task 4: Compare the CONCATENATE function and ampersand (&) operator.

📸 **Answer:** Both columns return the same result in every row. The & operator is shorter and easier to read. CONCATENATE is the older function (newer versions offer CONCAT and TEXTJOIN).

Task 5: Explain when to use CONCATENATE vs ampersand (&).

📸 **Answer:** Use & for quick, simple joins. Use CONCATENATE (or CONCAT and TEXTJOIN) for compatibility with older files or when joining many cells or a range with a delimiter.

6.7 Using Find & Replace Feature

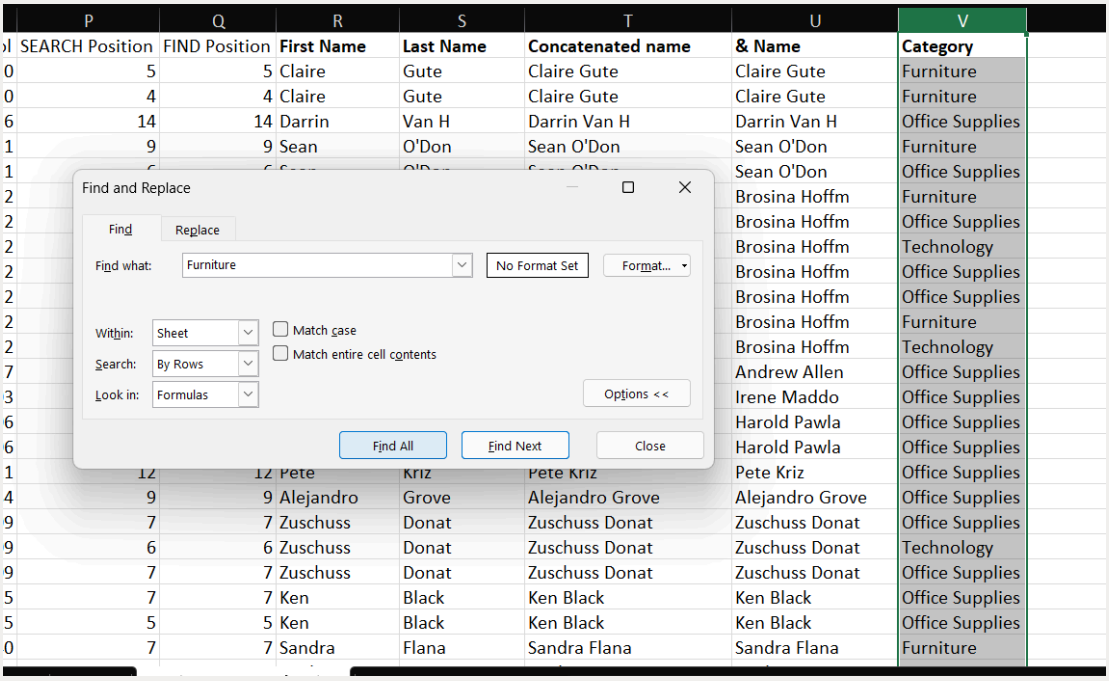
Task 1: Create a dataset with various instances of specific text (e.g., product codes).

📸 **Answer:** The Category column contains repeated text values: Furniture, Office Supplies and Technology (about 10,000 rows).

Task 2: Use the Find feature to locate all instances of a specific text.

📸 **Answer:** Ctrl+F → Find what: Furniture → Find All lists every cell containing "Furniture".

Screenshot:

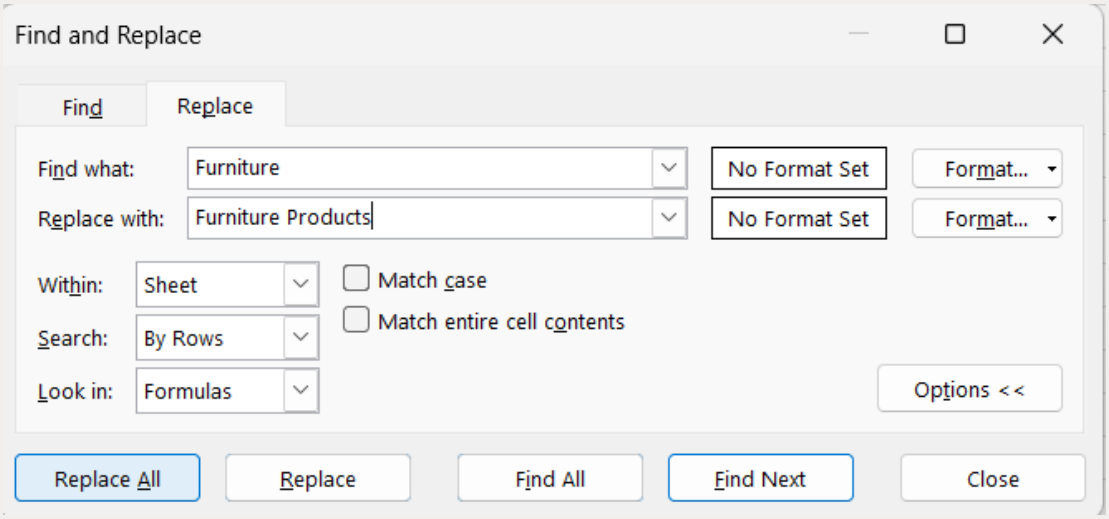


P	Q	R	S	T	U	V
SEARCH Position	FIND Position	First Name	Last Name	Concatenated name	& Name	Category
5	5	Claire	Gute	Claire Gute	Claire Gute	Furniture
4	4	Claire	Gute	Claire Gute	Claire Gute	Furniture
14	14	Darrin	Van H	Darrin Van H	Darrin Van H	Office Supplies
9	9	Sean	O'Don	Sean O'Don	Sean O'Don	Furniture
					Sean O'Don	Office Supplies
					Brosina Hoffm	Furniture
					Brosina Hoffm	Office Supplies
					Brosina Hoffm	Technology
					Brosina Hoffm	Office Supplies
					Brosina Hoffm	Office Supplies
					Brosina Hoffm	Furniture
					Brosina Hoffm	Technology
					Andrew Allen	Office Supplies
					Irene Maddo	Office Supplies
					Harold Pawla	Office Supplies
					Harold Pawla	Office Supplies
12	12	Pete	Kriz	Pete Kriz	Pete Kriz	Office Supplies
9	9	Alejandro	Grove	Alejandro Grove	Alejandro Grove	Office Supplies
7	7	Zuschuss	Donat	Zuschuss Donat	Zuschuss Donat	Office Supplies
6	6	Zuschuss	Donat	Zuschuss Donat	Zuschuss Donat	Technology
7	7	Zuschuss	Donat	Zuschuss Donat	Zuschuss Donat	Office Supplies
7	7	Ken	Black	Ken Black	Ken Black	Office Supplies
5	5	Ken	Black	Ken Black	Ken Black	Office Supplies
7	7	Sandra	Flana	Sandra Flana	Sandra Flana	Furniture

Task 3: Apply the Replace feature to change all instances of a specific text.

Answer: Ctrl+H → Find what: Furniture → Replace with: Furniture Products → Replace All. Excel confirmed "We made 2121 replacements".

Screenshot:



Task 4: Experiment with finding and replacing text using wildcards.

Answer: Use `*` for any number of characters and `?` for one. For example, find `Off*Supplies` to match "Office Supplies", or `Tech*` to match "Technology". Tick "Match entire cell contents" for precise replacements.

Task 5: Explain the benefits of using Find & Replace for data cleaning.

Answer: It changes thousands of entries at once, fixes typos and standardises naming. It is much faster and less error-prone than editing cells by hand.

Section 7: Pivot Table A-2-Z

7.1 Getting Started with Pivot Table

Task 1: Create a dataset with sales data (e.g., product, date, sales amount).

📸 **Answer:** The Orders sheet (Order Date, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit) is the source data.

Task 2: Insert a Pivot Table to analyze the sales data.

📸 **Answer:** Insert → PivotTable → select the Orders table → New Worksheet. This creates the "Pivot 7 Sales Analysis" sheet.

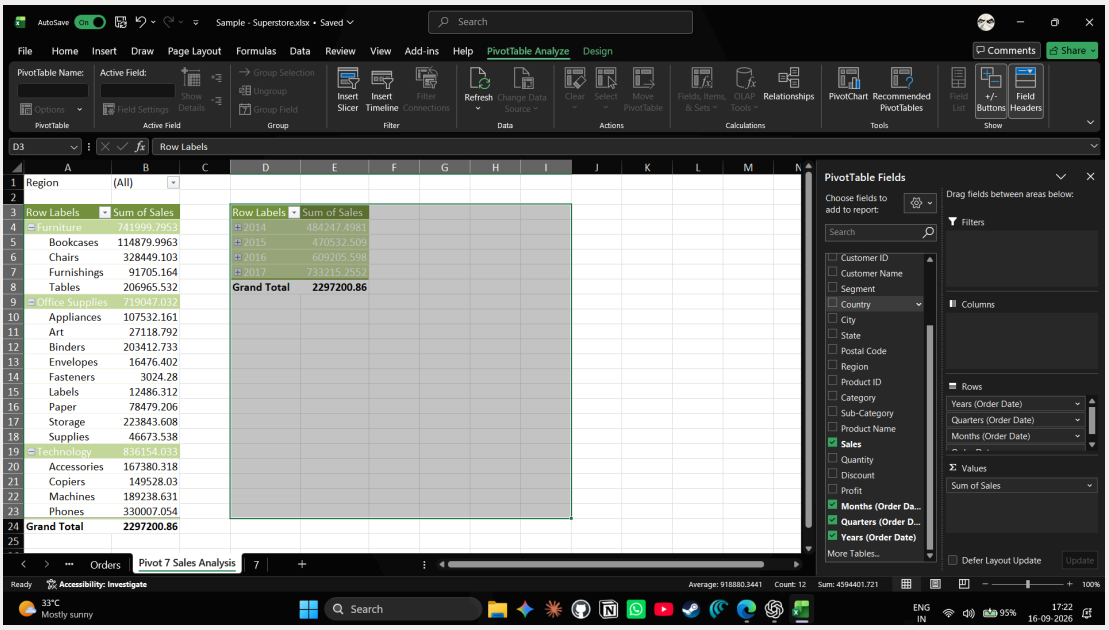
Task 3: Add fields to the Pivot Table for analysis (e.g., product, sales amount).

📸 **Answer:** Rows: Category then Sub-Category. Values: Sum of Sales. Total sales is **22,97,200.86**. Technology leads with 8,36,154. Furniture is 7,41,999.80 and Office Supplies 7,19,047.03.

Task 4: Apply filters to the Pivot Table to refine the data analysis.

📸 **Answer:** A Region filter (currently "All") sits above the pivot. Pick a region to see only its sales. Row Labels also have filter drop-downs.

Screenshot:



Task 5: Explain the basic components and functionalities of a Pivot Table.

📸 **Answer:** A PivotTable has four areas: Filters (restrict data), Columns, Rows (categories) and Values (calculations such as sum, count and average). It summarises large data quickly without formulas and updates on Refresh.

7.2 Grouping and Multiple Tables in Pivot Table

Task 1: Use the Pivot Table to group sales data by month.

📸 **Answer:** Order Date is grouped into Years, Quarters and Months (Rows area shows all three).

Screenshot:

Task 2: Create multiple tables within the Pivot Table for different analyses.

📸 **Answer:** Two pivots on one sheet: Category / Sub-Category sales on the left and Year / Quarter / Month sales on the right. Both total 22,97,200.86.

Screenshot:

Task 3: Experiment with grouping data by different periods (e.g., quarter, year).

📸 **Answer:** Yearly sales: 2014 = 4,84,247; 2015 = 4,70,533; 2016 = 6,09,206; 2017 = 7,33,215. Expand a year to see its quarters and months.

Screenshot:

Task 4: Use multiple tables to compare sales across different products.

📸 **Answer:** Top sub-categories by sales: Phones 3,30,007; Chairs 3,28,449; Storage 2,23,844; Tables 2,06,966; Binders 2,03,413. Fasteners is the lowest at 3,024.

Screenshot:

Task 5: Explain the benefits of grouping and using multiple tables in Pivot Tables.

📸 **Answer:** Grouping reveals trends over time periods, and multiple pivots on one sheet let you view the same data from different angles side by side without rebuilding formulas.

Screenshot:

7.3 Pivot Table for Grouping Via Date, Blank Cells

Task 1: Group sales data by date in the Pivot Table.

📸 **Answer:** Right-click a date in the pivot → Group → select Years, Quarters and Months. The pivot shows 2014 to 2017 with expandable sub-levels.

Screenshot:

Task 2: Handle blank cells in the Pivot Table.

📸 **Answer:** A Region pivot shows a "(blank)" row with sales of 660. Fill the blanks (for example with "Unknown" using Go To Special) or filter the blank out, then Refresh.

Screenshot:

Task 3: Analyze the impact of blank cells on data grouping.

📸 **Answer:** Blank regions form their own "(blank)" group worth 660 of 20,945.16 total (about 3.2%). Region totals (Central 4,032; East 5,173; South 3,890; West 7,190) therefore understate the true figures.

Screenshot:

Task 4: Use different date formats for grouping data.

📸 **Answer:** Group by Years, Quarters or Months, or by Days with a number of days. Row labels show them as 2014, Qtr1 and Jan.

Screenshot:

Task 5: Explain the importance of managing blank cells in Pivot Table analysis.

📸 **Answer:** Blanks create stray "(blank)" groups, distort totals and break date grouping (Excel cannot group a date field that contains blanks). Cleaning them first keeps results accurate.

Screenshot:

7.4 Custom Grouping, Conditional Formatting in Pivot Table

Task 1: Create custom groups in the Pivot Table (e.g., grouping products by category).

📸 **Answer:** Select Furniture and Office Supplies → Group → rename the group "Office Products" (1,461,046.83). Technology stays on its own at 8,36,154.03.

Screenshot:

Task 2: Apply conditional formatting to highlight key data points.

📸 **Answer:** A colour scale on Sum of Sales highlights the highest and lowest values in red-to-blue shades.

Screenshot:

Task 3: Experiment with different conditional formatting rules.

📸 **Answer:** Tried two rules: a colour scale and data bars. Icon sets and Top 10 rules work the same way (Home → Conditional Formatting).

Screenshot:

Task 4: Use custom grouping to create a more detailed data analysis.

📸 **Answer:** Grouping the categories into "Office Products" versus Technology gives a two-tier comparison: 14.6 lakh against 8.4 lakh. Furniture and Office Supplies stay visible underneath.

Screenshot:

Task 5: Explain how custom grouping and conditional formatting enhance Pivot Table analysis.

📸 **Answer:** Custom groups organise related items into meaningful business groups. Conditional formatting makes high and low values visible at a glance, so insights are faster to read and present.

Screenshot:

Section 8: Everything about LOOKUP Formulas

8.1 Basics of VLOOKUP

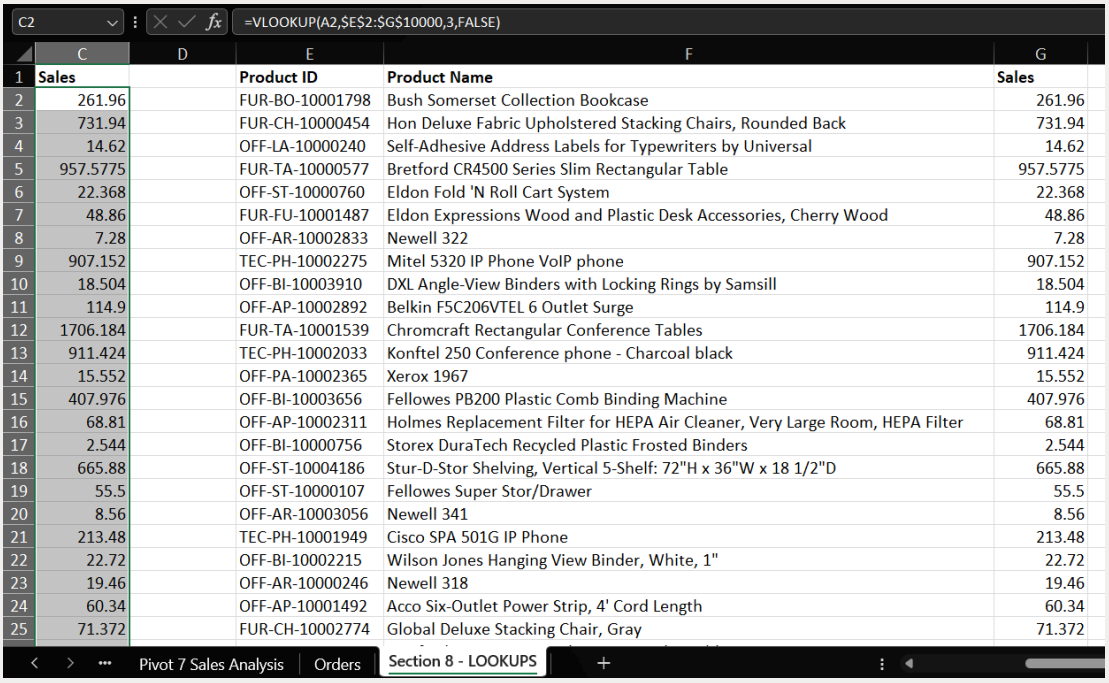
Task 1: Create a dataset with product information (e.g., product ID, name, price).

📸 **Answer:** A lookup table of Product ID, Product Name and Sales (used as the price) in columns E to G, and the lookup list of Product IDs in column A.

Task 2: Use the VLOOKUP function to find the price of a product based on its ID.

Answer: Name: `=VLOOKUP(A2,$E$2:$G$10000,2,FALSE)` gives "Bush Somerset Collection Bookcase". Price/Sales: `=VLOOKUP(A2,$E$2:$G$10000,3,FALSE)` gives 261.96.

Screenshot:

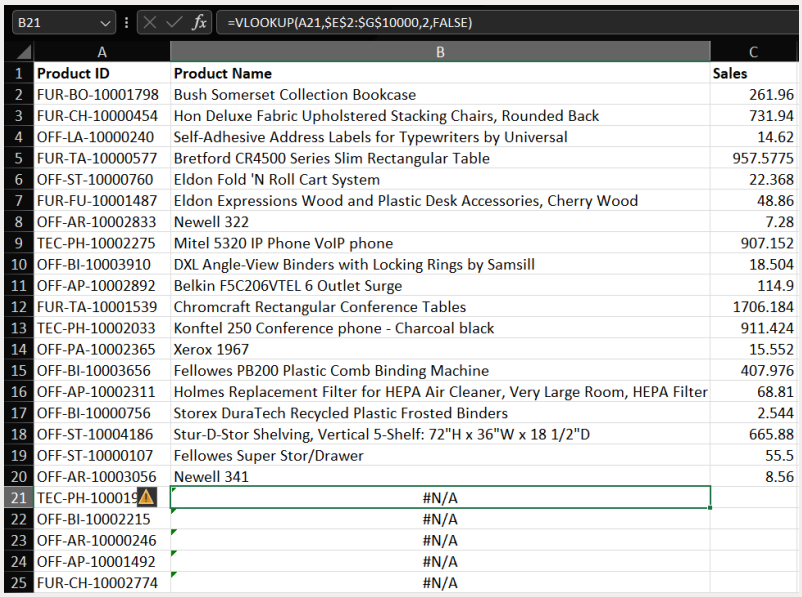


	C	D	E	F	G
1	Sales		Product ID	Product Name	Sales
2	261.96		FUR-BO-10001798	Bush Somerset Collection Bookcase	261.96
3	731.94		FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94
4	14.62		OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal	14.62
5	957.5775		FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	957.5775
6	22.368		OFF-ST-10000760	Eldon Fold 'N Roll Cart System	22.368
7	48.86		FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86
8	7.28		OFF-AR-10002833	Newell 322	7.28
9	907.152		TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	907.152
10	18.504		OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Samsill	18.504
11	114.9		OFF-AP-10002892	Belkin F5C206VTEL 6 Outlet Surge	114.9
12	1706.184		FUR-TA-10001539	Chromcraft Rectangular Conference Tables	1706.184
13	911.424		TEC-PH-10002033	Konftel 250 Conference phone - Charcoal black	911.424
14	15.552		OFF-PA-10002365	Xerox 1967	15.552
15	407.976		OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine	407.976
16	68.81		OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81
17	2.544		OFF-BI-10000756	Storex DuraTech Recycled Plastic Frosted Binders	2.544
18	665.88		OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88
19	55.5		OFF-ST-10000107	Fellowes Super Stor/Drawer	55.5
20	8.56		OFF-AR-10003056	Newell 341	8.56
21	213.48		TEC-PH-10001949	Cisco SPA 501G IP Phone	213.48
22	22.72		OFF-BI-10002215	Wilson Jones Hanging View Binder, White, 1"	22.72
23	19.46		OFF-AR-10000246	Newell 318	19.46
24	60.34		OFF-AP-10001492	Acco Six-Outlet Power Strip, 4' Cord Length	60.34
25	71.372		FUR-CH-10002774	Global Deluxe Stacking Chair, Gray	71.372

Task 3: Experiment with different lookup values in the VLOOKUP function.

Answer: Copying the formula down looks up each ID, for example FUR-CH-10000454 gives 731.94 and OFF-LA-10000240 gives 14.62. The test ID "TEST" returns an error.

Screenshot:

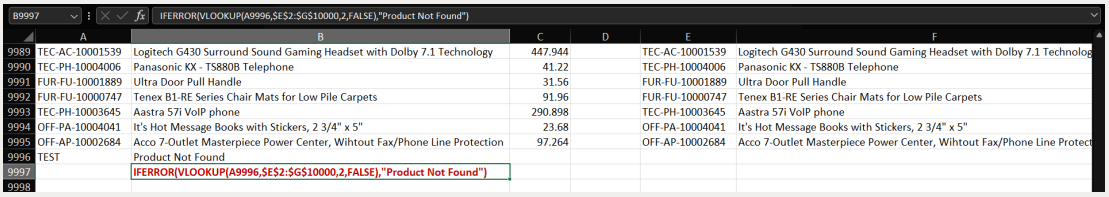


	A	B	C
1	**Product ID**	**Product Name**	**Sales**
2	FUR-BO-10001798	Bush Somerset Collection Bookcase	261.96
3	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94
4	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal	14.62
5	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	957.5775
6	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	22.368
7	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86
8	OFF-AR-10002833	Newell 322	7.28
9	TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	907.152
10	OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Samsill	18.504
11	OFF-AP-10002892	Belkin F5C206VTEL 6 Outlet Surge	114.9
12	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	1706.184
13	TEC-PH-10002033	Konftel 250 Conference phone - Charcoal black	911.424
14	OFF-PA-10002365	Xerox 1967	15.552
15	OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine	407.976
16	OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81
17	OFF-BI-10000756	Storex DuraTech Recycled Plastic Frosted Binders	2.544
18	OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88
19	OFF-ST-10000107	Fellowes Super Stor/Drawer	55.5
20	OFF-AR-10003056	Newell 341	8.56
21	TEC-PH-100019	#N/A	
22	OFF-BI-10002215	#N/A	
23	OFF-AR-10000246	#N/A	
24	OFF-AP-10001492	#N/A	
25	FUR-CH-10002774	#N/A	

Task 4: Handle errors in VLOOKUP results using IFERROR.

Answer: `=IFERROR(VLOOKUP(A9996,$E$2:$G$10000,2,FALSE),"Product Not Found")` returns "Product Not Found" for the ID "TEST" instead of #N/A.

Screenshot:



	A	B	C	D	E	F
9989	TEC-AC-10001539	Logitech G430 Surround Sound Gaming Headset with Dolby 7.1 Technology	447.944		TEC-AC-10001539	Logitech G430 Surround Sound Gaming Headset with Dolby 7.1 Technolog
9990	TEC-PH-10004006	Panasonic KX - TS880B Telephone	41.22		TEC-PH-10004006	Panasonic KX - TS880B Telephone
9991	FUR-FU-10001889	Ultra Door Pull Handle	31.56		FUR-FU-10001889	Ultra Door Pull Handle
9992	FUR-FU-10000747	Tenex B1-RE Series Chair Mats for Low Pile Carpets	91.96		FUR-FU-10000747	Tenex B1-RE Series Chair Mats for Low Pile Carpets
9993	TEC-PH-10003645	Aastra 57i VoIP phone	290.898		TEC-PH-10003645	Aastra 57i VoIP phone
9994	OFF-PA-10004041	It's Hot Message Books with Stickers, 2 3/4" x 5"	23.68		OFF-PA-10004041	It's Hot Message Books with Stickers, 2 3/4" x 5"
9995	OFF-AP-10002684	Acco 7-Outlet Masterpiece Power Center, Wihtout Fax/Phone Line Protection	97.264		OFF-AP-10002684	Acco 7-Outlet Masterpiece Power Center, Wihtout Fax/Phone Line Protect
9996	TEST	Product Not Found				
9997		IFERROR(VLOOKUP(A9996,$E$2:$G$10000,2,FALSE),"Product Not Found")				
9998						

Task 5: Explain the basic syntax and functionality of the VLOOKUP function.

Answer: `VLOOKUP(lookup_value, table_array, col_index_num, range_lookup)`. It searches the first column of the table for the value and returns the value

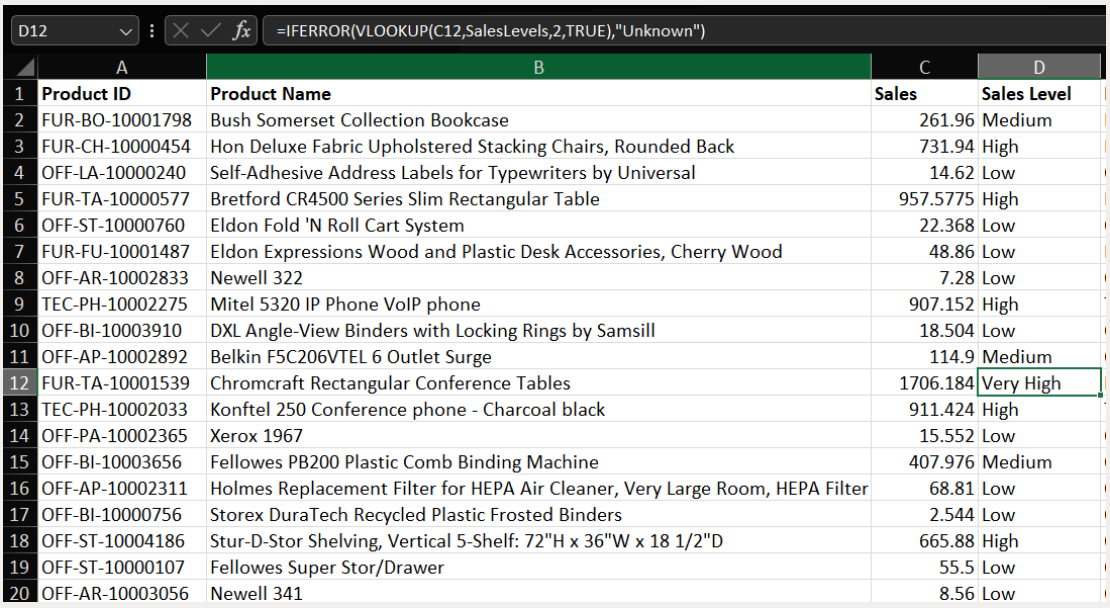
from the chosen column in the same row. FALSE means exact match. The lookup value must be in the leftmost column.

8.2 More about VLOOKUP

Task 1: Use VLOOKUP with approximate match (TRUE) to find data in a range.

📸 **Answer:** `=VLOOKUP(C2,SalesLevels,2,TRUE)` assigns a Sales Level using the band table: 0 Low, 100 Medium, 500 High, 1000 Very High, 5000 Excellent. For example 261.96 gives Medium and 1706.18 gives Very High.

Screenshot:



D12 =IFERROR(VLOOKUP(C12,SalesLevels,2,TRUE),"Unknown")

	A	B	C	D
1	**Product ID**	**Product Name**	**Sales**	**Sales Level**
2	FUR-BO-10001798	Bush Somerset Collection Bookcase	261.96	Medium
3	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	High
4	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal	14.62	Low
5	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	957.5775	High
6	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	22.368	Low
7	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	Low
8	OFF-AR-10002833	Newell 322	7.28	Low
9	TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	907.152	High
10	OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Samsill	18.504	Low
11	OFF-AP-10002892	Belkin F5C206VTEL 6 Outlet Surge	114.9	Medium
12	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	1706.184	Very High
13	TEC-PH-10002033	Konftel 250 Conference phone - Charcoal black	911.424	High
14	OFF-PA-10002365	Xerox 1967	15.552	Low
15	OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine	407.976	Medium
16	OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	Low
17	OFF-BI-10000756	Storex DuraTech Recycled Plastic Frosted Binders	2.544	Low
18	OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	High
19	OFF-ST-10000107	Fellowes Super Stor/Drawer	55.5	Low
20	OFF-AR-10003056	Newell 341	8.56	Low

Task 2: Compare exact match (FALSE) vs approximate match (TRUE) in VLOOKUP.

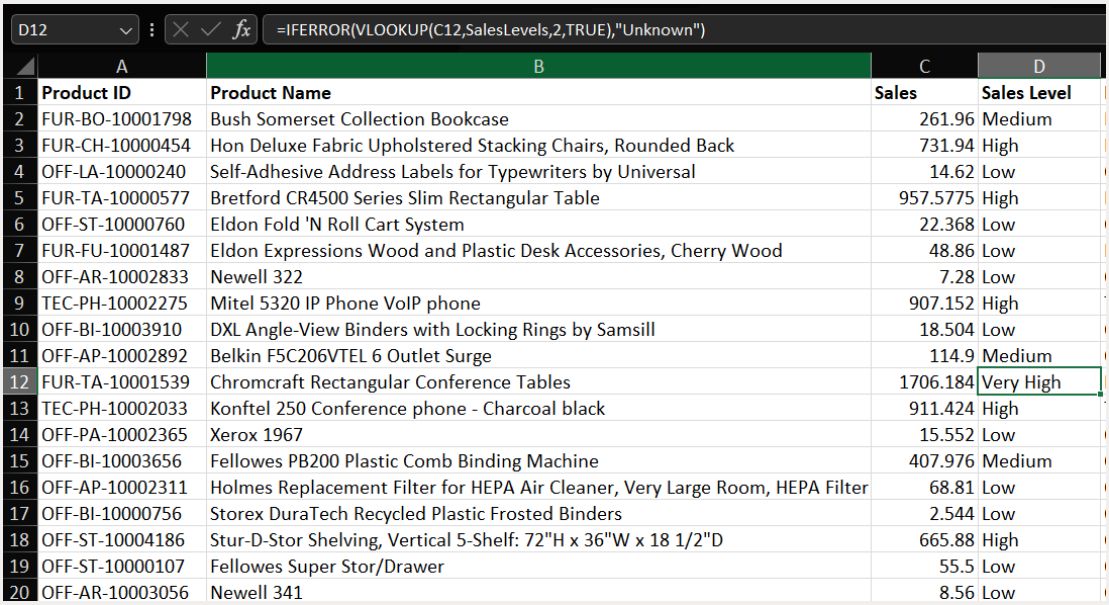
📸 **Answer:** Exact (FALSE) needs the value to exist (a product ID). Approximate (TRUE) finds the largest value that does not exceed the

lookup value, so 731.94 falls in the 500 band and returns High. Approximate needs the first column sorted ascendiNG.

Task 3: Use named ranges with VLOOKUP for better readability.

Answer: The band table is named `SalesLevels` (Formulas → Define Name), so the formula reads `VLOOKUP(C2,SalesLevels,2,TRUE)` instead of cell addresses.

Screenshot:

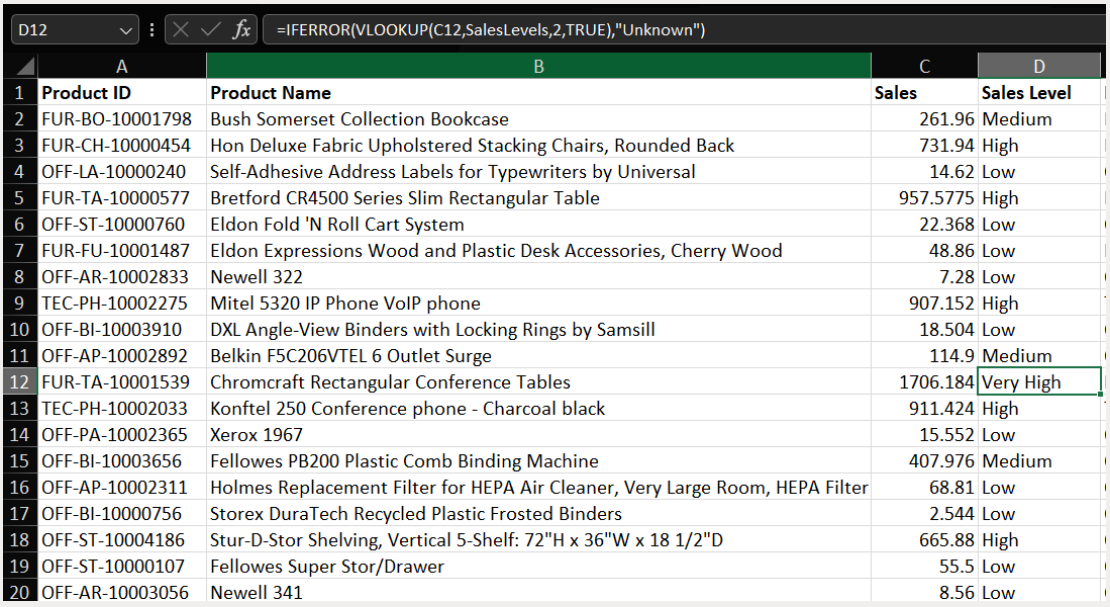


	A	B	C	D
1	**Product ID**	**Product Name**	**Sales**	**Sales Level**
2	FUR-BO-10001798	Bush Somerset Collection Bookcase	261.96	Medium
3	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	High
4	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal	14.62	Low
5	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	957.5775	High
6	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	22.368	Low
7	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	Low
8	OFF-AR-10002833	Newell 322	7.28	Low
9	TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	907.152	High
10	OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Samsill	18.504	Low
11	OFF-AP-10002892	Belkin F5C206VTEL 6 Outlet Surge	114.9	Medium
12	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	1706.184	Very High
13	TEC-PH-10002033	Konftel 250 Conference phone - Charcoal black	911.424	High
14	OFF-PA-10002365	Xerox 1967	15.552	Low
15	OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine	407.976	Medium
16	OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	Low
17	OFF-BI-10000756	Storex DuraTech Recycled Plastic Frosted Binders	2.544	Low
18	OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	High
19	OFF-ST-10000107	Fellowes Super Stor/Drawer	55.5	Low
20	OFF-AR-10003056	Newell 341	8.56	Low

Task 4: Combine VLOOKUP with other functions for advanced data retrieval.

Answer: `=IFERROR(VLOOKUP(C2,SalesLevels,2,TRUE),"Unknown")` wraps VLOOKUP in IFERROR so a missing match shows "Unknown". It also works with MATCH or IF.

Screenshot:



D12 =IFERROR(VLOOKUP(C12,SalesLevels,2,TRUE),"Unknown")

	A	B	C	D
1	**Product ID**	**Product Name**	**Sales**	**Sales Level**
2	FUR-BO-10001798	Bush Somerset Collection Bookcase	261.96	Medium
3	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	High
4	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal	14.62	Low
5	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	957.5775	High
6	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	22.368	Low
7	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	Low
8	OFF-AR-10002833	Newell 322	7.28	Low
9	TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	907.152	High
10	OFF-BI-10003910	DXL Angle-View Binders with Locking Rings by Samsill	18.504	Low
11	OFF-AP-10002892	Belkin F5C206VTEL 6 Outlet Surge	114.9	Medium
12	FUR-TA-10001539	Chromcraft Rectangular Conference Tables	1706.184	Very High
13	TEC-PH-10002033	Konftel 250 Conference phone - Charcoal black	911.424	High
14	OFF-PA-10002365	Xerox 1967	15.552	Low
15	OFF-BI-10003656	Fellowes PB200 Plastic Comb Binding Machine	407.976	Medium
16	OFF-AP-10002311	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	Low
17	OFF-BI-10000756	Storex DuraTech Recycled Plastic Frosted Binders	2.544	Low
18	OFF-ST-10004186	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	High
19	OFF-ST-10000107	Fellowes Super Stor/Drawer	55.5	Low
20	OFF-AR-10003056	Newell 341	8.56	Low

Task 5: Explain when to use approximate match vs exact match in VLOOKUP.

📸 **Answer:** Use exact match for IDs, names and codes. Use approximate match for ranges and bands such as grades, tax slabs, commission tiers and sales levels (with sorted data).

8.3 VLOOKUP vs. HLOOKUP

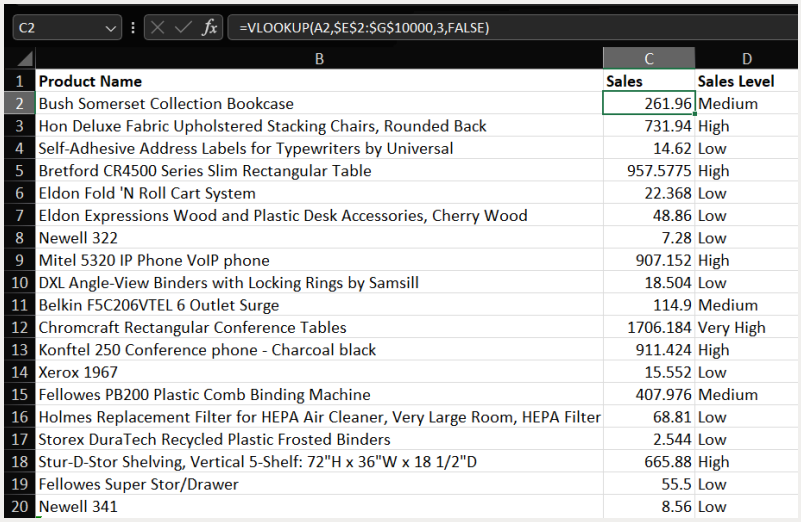
Task 1: Create a dataset suitable for both VLOOKUP and HLOOKUP.

📸 **Answer:** The Product ID / Name / Sales table runs vertically (for VLOOKUP). The monthly sales table (Month in one row, Sales below, Jan to Dec) runs horizontally (for HLOOKUP).

Task 2: Use VLOOKUP to retrieve data from a vertical dataset.

Answer: `=VLOOKUP(A2,$E$2:$G$10000,3,FALSE)` returns Sales for a Product ID, for example 261.96.

Screenshot:

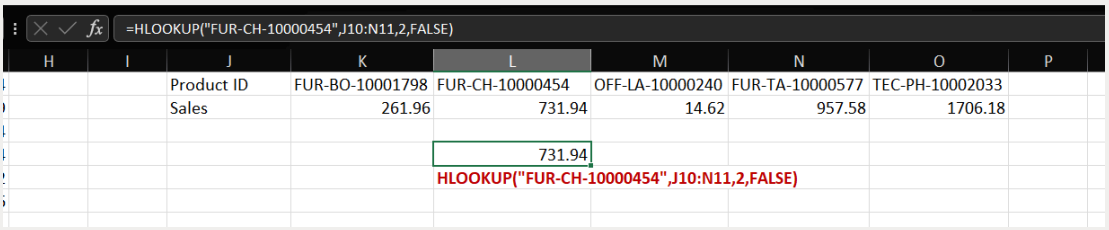


	B	C	D
1	**Product Name**	**Sales**	**Sales Level**
2	Bush Somerset Collection Bookcase	261.96	Medium
3	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	High
4	Self-Adhesive Address Labels for Typewriters by Universal	14.62	Low
5	Bretford CR4500 Series Slim Rectangular Table	957.5775	High
6	Eldon Fold 'N Roll Cart System	22.368	Low
7	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	Low
8	Newell 322	7.28	Low
9	Mitel 5320 IP Phone VoIP phone	907.152	High
10	DXL Angle-View Binders with Locking Rings by Samsill	18.504	Low
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12	Chromcraft Rectangular Conference Tables	1706.184	Very High
13	Konftel 250 Conference phone - Charcoal black	911.424	High
14	Xerox 1967	15.552	Low
15	Fellowes PB200 Plastic Comb Binding Machine	407.976	Medium
16	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	Low
17	Storex DuraTech Recycled Plastic Frosted Binders	2.544	Low
18	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	High
19	Fellowes Super Stor/Drawer	55.5	Low
20	Newell 341	8.56	Low

Task 3: Use HLOOKUP to retrieve data from a horizontal dataset.

Answer: `=HLOOKUP("Jul",G1:S2,2,FALSE)` returns July sales of 27,500 from the horizontal Month and Sales rows.

Screenshot:



Task 4: Compare the syntax and usage of VLOOKUP vs HLOOKUP.

 **Answer:** Both use (lookup_value, table_array, index, range_lookup). VLOOKUP searches the first column and returns a column number. HLOOKUP searches the first row and returns a row number.

Task 5: Explain the scenarios where VLOOKUP or HLOOKUP would be more appropriate.

 **Answer:** VLOOKUP suits data in columns (customer, product and employee lists). HLOOKUP suits data with headers across the top, such as monthly or yearly layouts. In modern Excel, XLOOKUP handles both.

Section 9: Conditional Aggregation

9.1 Basics of COUNTIF, SUMIF, AVERAGEIF Formula

Task 1: Create a dataset with sales data (e.g., product, sales amount).

 **Answer:** Dataset with Region, Category and Sales columns (South, West, Central, East and Furniture, Office Supplies, Technology).

Task 2: Use COUNTIF to count the number of sales above a certain amount.

 **Answer:** `=COUNTIF(D:D,">500")` gives **8** sales above 500.

Task 3: Apply SUMIF to calculate the total sales for a specific product.

 **Answer:** `=SUMIF(C:C,F6,D:D)` with F6 = Furniture gives **7,905.95**.

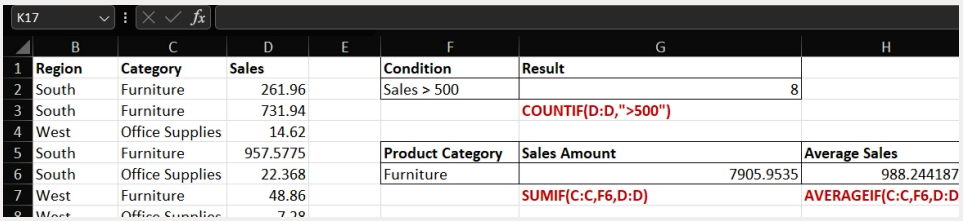
Task 4: Use AVERAGEIF to find the average sales amount for a certain product category.

📸 **Answer:** `=AVERAGEIF(C:C,F6,D:D)` gives **988.24** for Furniture.

Task 5: Compare COUNTIF, SUMIF, and AVERAGEIF formulas and their use cases.

📸 **Answer:** COUNTIF counts how many cells meet a condition (8 sales above 500). SUMIF adds values that meet it (7,905.95 for Furniture). AVERAGEIF averages them (988.24). SUMIF divided by the count of Furniture rows (8) equals AVERAGEIF.

Screenshot:



	B	C	D	E	F	G	H
1	Region	Category	Sales		Condition	Result	
2	South	Furniture	261.96		Sales > 500	8	
3	South	Furniture	731.94			COUNTIF(D:D,">500")	
4	West	Office Supplies	14.62				
5	South	Furniture	957.5775		Product Category	Sales Amount	Average Sales
6	South	Office Supplies	22.368		Furniture	7905.9535	988.244187
7	West	Furniture	48.86			SUMIF(C:C,F6,D:D)	AVERAGEIF(C:C,F6,D:D

9.2 COUNTIFS & SUMIFS

Task 1: Create a dataset with multiple criteria (e.g., product, region, sales amount).

📸 **Answer:** Same dataset with Region, Category and Sales, so multiple conditions can be combined.

Task 2: Use COUNTIFS to count the number of sales meeting multiple criteria.

📸 **Answer:** `=COUNTIFS(B:B,"West",D:D,">500")` gives **4** West sales above 500.

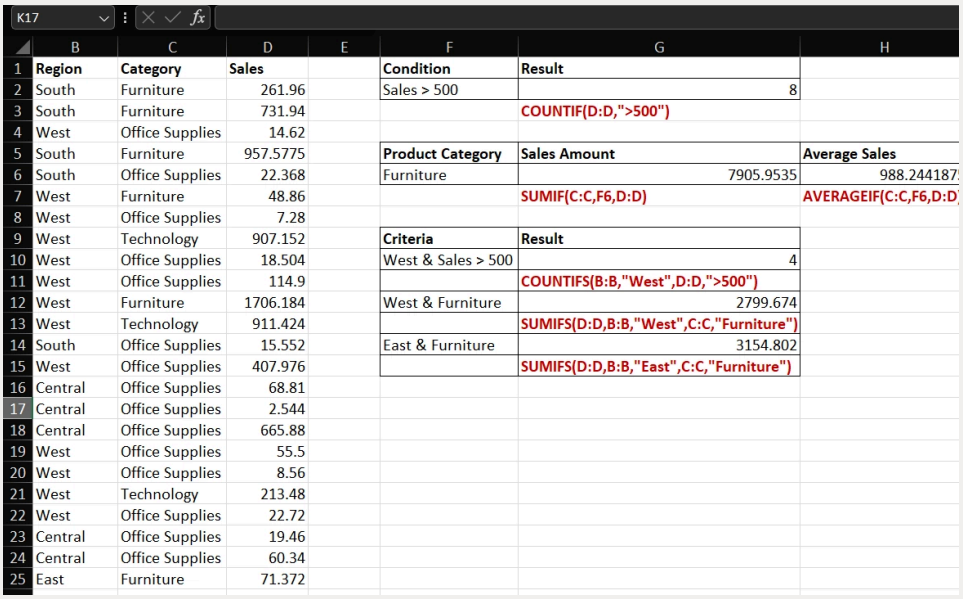
Task 3: Apply SUMIFS to calculate total sales meeting multiple criteria.

Answer: `=SUMIFS(D:D,B:B,"West",C:C,"Furniture")` gives **2,799.67** for West Furniture.

Task 4: Experiment with different combinations of criteria in COUNTIFS and SUMIFS.

Answer: Changing the region gives `=SUMIFS(D:D,B:B,"East",C:C,"Furniture")` = **3,154.80** for East Furniture. Other combinations (Central and Office Supplies, for example) work the same way.

Screenshot:



	B	C	D	E	F	G	H
1	Region	Category	Sales		Condition	Result	
2	South	Furniture	261.96		Sales > 500	8	
3	South	Furniture	731.94			COUNTIF(D:D,">500")	
4	West	Office Supplies	14.62				
5	South	Furniture	957.5775		Product Category	Sales Amount	Average Sales
6	South	Office Supplies	22.368		Furniture	7905.9535	988.244187
7	West	Furniture	48.86			SUMIF(C:C,F6,D:D)	AVERAGEIF(C:C,F6,D:D)
8	West	Office Supplies	7.28				
9	West	Technology	907.152		Criteria	Result	
10	West	Office Supplies	18.504		West & Sales > 500	4	
11	West	Office Supplies	114.9			COUNTIFS(B:B,"West",D:D,">500")	
12	West	Furniture	1706.184		West & Furniture	2799.674	
13	West	Technology	911.424			SUMIFS(D:D,B:B,"West",C:C,"Furniture")	
14	South	Office Supplies	15.552		East & Furniture	3154.802	
15	West	Office Supplies	407.976			SUMIFS(D:D,B:B,"East",C:C,"Furniture")	
16	Central	Office Supplies	68.81				
17	Central	Office Supplies	2.544				
18	Central	Office Supplies	665.88				
19	West	Office Supplies	55.5				
20	West	Office Supplies	8.56				
21	West	Technology	213.48				
22	West	Office Supplies	22.72				
23	Central	Office Supplies	19.46				
24	Central	Office Supplies	60.34				
25	East	Furniture	71.372				

Task 5: Explain the importance of using COUNTIFS and SUMIFS for complex data analysis.

📸 **Answer:** COUNTIFS counts records that meet several conditions, and SUMIFS adds values that meet several conditions. They answer real business questions (such as West Furniture sales) without filters or pivots.

Section 10: Conditional Formatting

10.1 Conditional Formatting - Color Cells Using Conditions

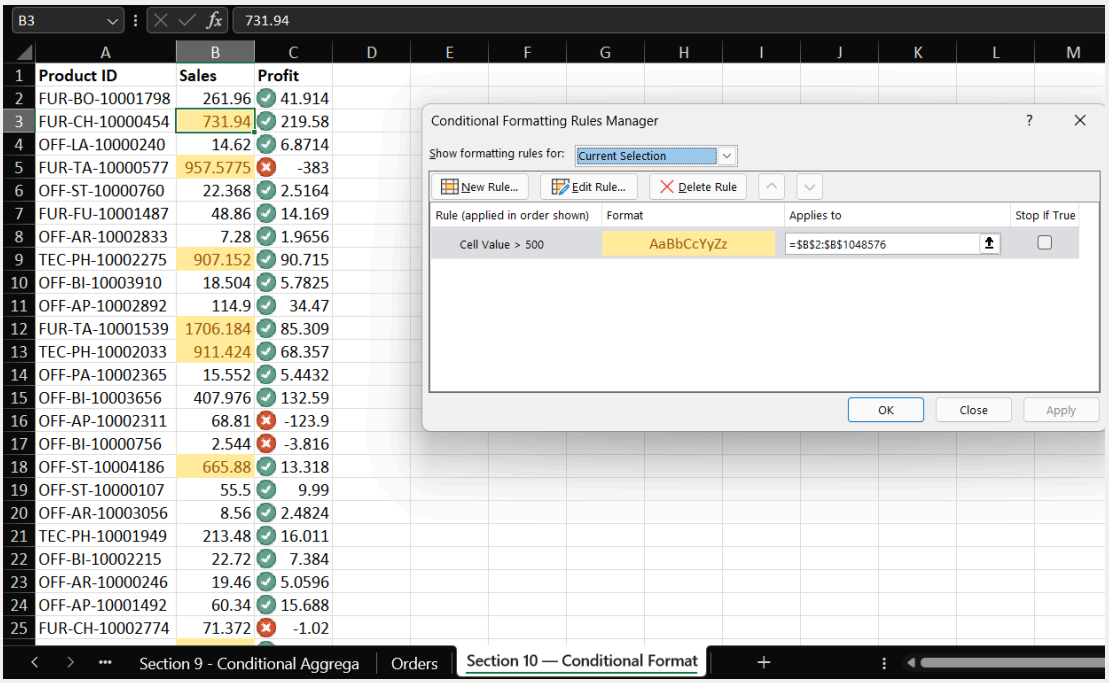
Task 1: Create a dataset with sales data (e.g., product, sales amount).

📸 **Answer:** Dataset with Product ID, Sales and Profit columns (for example FUR-BO-10001798, 261.96, 41.91).

Task 2: Apply conditional formatting to highlight sales greater than a specified amount.

📸 **Answer:** Rules Manager shows "Cell Value > 500" applied to the Sales column with an orange fill. Highlighted: 731.94, 957.58, 907.15, 1706.18, 911.42 and 665.88.

Screenshot:



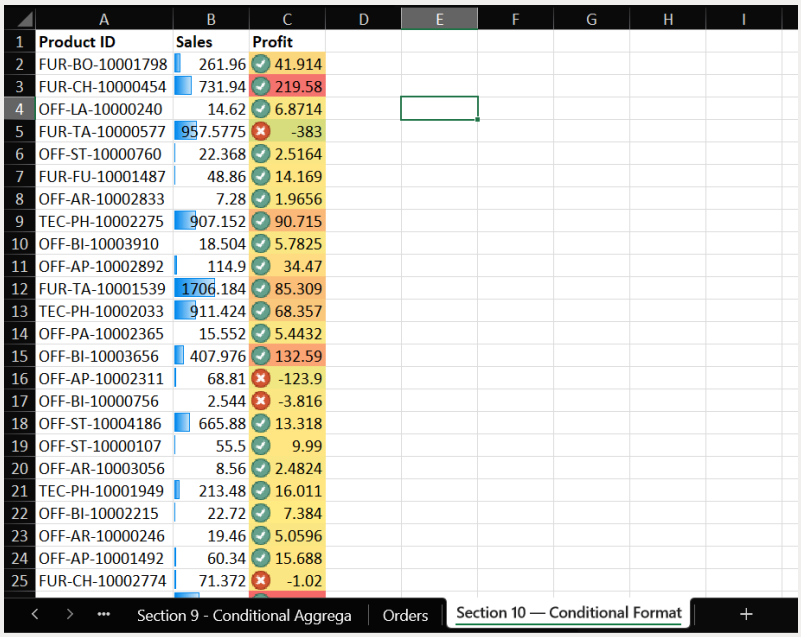
	A	B	C
1	Product ID	Sales	Profit
2	FUR-BO-10001798	261.96	41.914
3	FUR-CH-10000454	731.94	219.58
4	OFF-LA-10000240	14.62	6.8714
5	FUR-TA-10000577	957.5775	-383
6	OFF-ST-10000760	22.368	2.5164
7	FUR-FU-10001487	48.86	14.169
8	OFF-AR-10002833	7.28	1.9656
9	TEC-PH-10002275	907.152	90.715
10	OFF-BI-10003910	18.504	5.7825
11	OFF-AP-10002892	114.9	34.47
12	FUR-TA-10001539	1706.184	85.309
13	TEC-PH-10002033	911.424	68.357
14	OFF-PA-10002365	15.552	5.4432
15	OFF-BI-10003656	407.976	132.59
16	OFF-AP-10002311	68.81	-123.9
17	OFF-BI-10000756	2.544	-3.816
18	OFF-ST-10004186	665.88	13.318
19	OFF-ST-10000107	55.5	9.99
20	OFF-AR-10003056	8.56	2.4824
21	TEC-PH-10001949	213.48	16.011
22	OFF-BI-10002215	22.72	7.384
23	OFF-AR-10000246	19.46	5.0596
24	OFF-AP-10001492	60.34	15.688
25	FUR-CH-10002774	71.372	-1.02

Task 3: Use color scales to represent different ranges of sales amounts.

📸 **Answer:** A 3-colour scale on the Profit column shades low, middle and high values red, yellow and green.

Task 4: Experiment with data bars to visually represent sales data.

📸 **Answer:** Blue data bars on the Sales column show relative size at a glance. 1,706.18 has the longest bar.



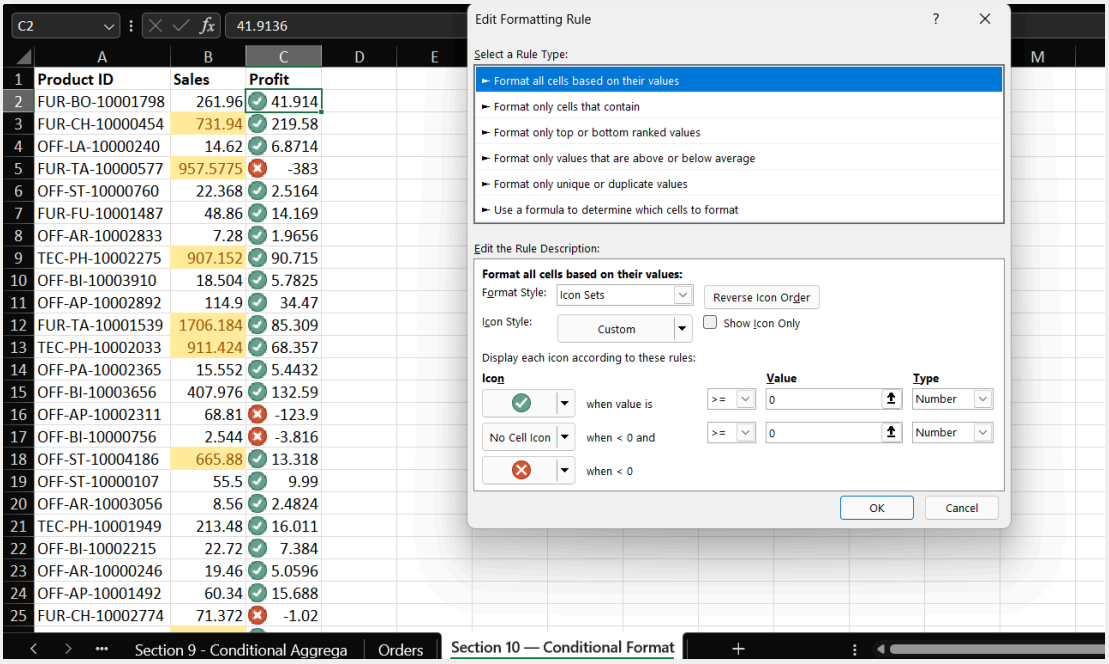
	A	B	C	D	E	F	G	H	I
1	Product ID	Sales	Profit						
2	FUR-BO-10001798	261.96	41.914						
3	FUR-CH-10000454	731.94	219.58						
4	OFF-LA-10000240	14.62	6.8714						
5	FUR-TA-10000577	957.5775	-383						
6	OFF-ST-10000760	22.368	2.5164						
7	FUR-FU-10001487	48.86	14.169						
8	OFF-AR-10002833	7.28	1.9656						
9	TEC-PH-10002275	907.152	90.715						
10	OFF-BI-10003910	18.504	5.7825						
11	OFF-AP-10002892	114.9	34.47						
12	FUR-TA-10001539	1706.184	85.309						
13	TEC-PH-10002033	911.424	68.357						
14	OFF-PA-10002365	15.552	5.4432						
15	OFF-BI-10003656	407.976	132.59						
16	OFF-AP-10002311	68.81	-123.9						
17	OFF-BI-10000756	2.544	-3.816						
18	OFF-ST-10004186	665.88	13.318						
19	OFF-ST-10000107	55.5	9.99						
20	OFF-AR-10003056	8.56	2.4824						
21	TEC-PH-10001949	213.48	16.011						
22	OFF-BI-10002215	22.72	7.384						
23	OFF-AR-10000246	19.46	5.0596						
24	OFF-AP-10001492	60.34	15.688						
25	FUR-CH-10002774	71.372	-1.02						

Section 9 - Conditional Aggrega | Orders | Section 10 — Conditional Format | +

Task 5: Apply icon sets to show trends and performance indicators in the data.

📸 **Answer:** A custom icon set on Profit: green check when profit is 0 or above, red cross when below 0 (for example -383 and -123.9).

Screenshot:



Section 11: Report Consolidation

11.1 Hidden Trick of Consolidation Using SUM for Multiple Sheets

Task 1: Create multiple sheets with monthly sales data.

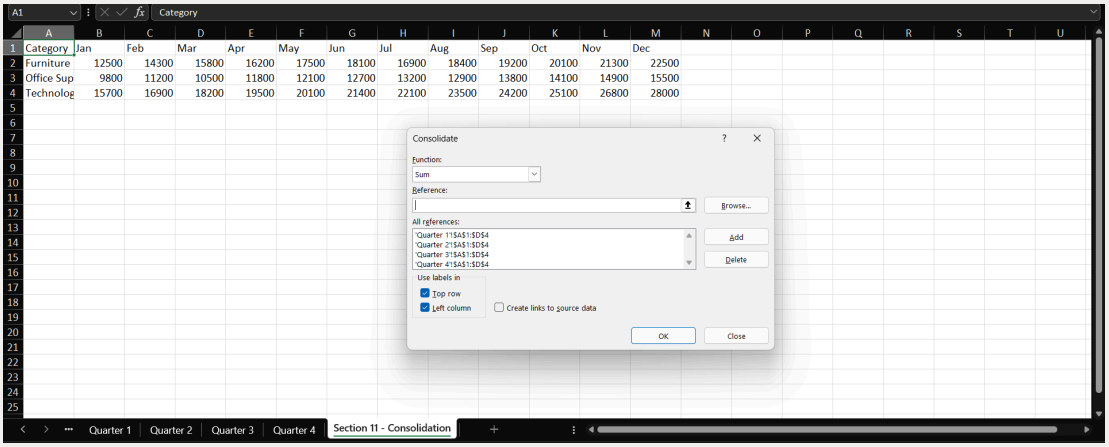
📸 **Answer:** Four sheets, Quarter 1 to Quarter 4, each with categories (Furniture, Office Supplies, Technology) and three months of sales. Together they cover Jan to Dec.

Task 2: Use the Consolidate feature with the SUM function to aggregate sales data from multiple sheets.

📸 **Answer:** Data → Consolidate → Function: Sum → add the range `A1:D4` from each Quarter sheet → tick Top row and Left column → OK. Result: Furniture Jan 12,500 through Dec 22,500, Office Supplies Jan 9,800 and

Technology Jan 15,700.

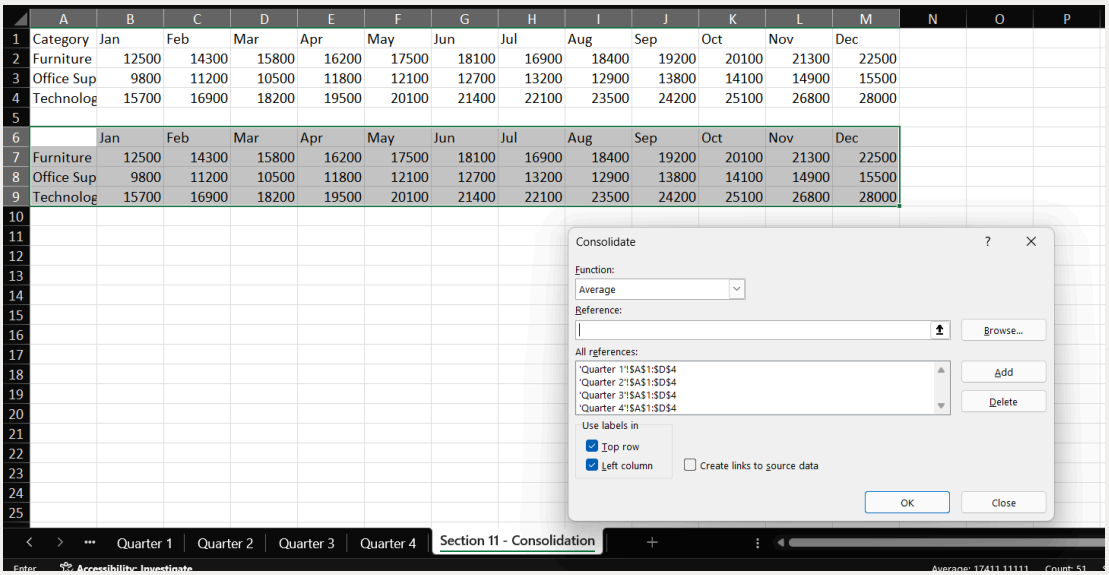
Screenshot:



Task 3: Experiment with different summary functions (e.g., Average, Count) in Consolidate.

📸 **Answer:** Changing Function to Average (or Count) re-runs the consolidation. Here Average gives the same figures as Sum, because each month appears in only one quarter sheet.

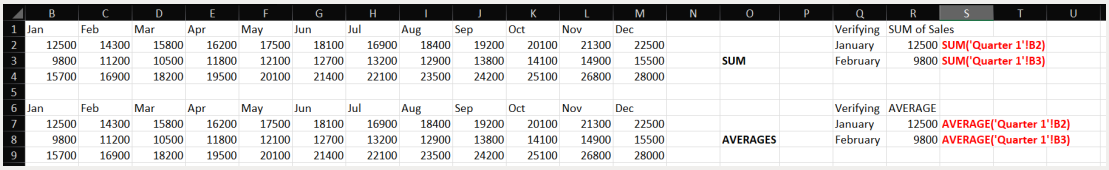
Screenshot:



Task 4: Verify the consolidated data for accuracy.

Answer: The consolidated values match the source sheets, for example Furniture Jan 12,500, Office Supplies Mar 10,500 and Technology Dec 28,000. Compare with the Quarter sheets or total with SUM.

Screenshot:



Task 5: Explain the benefits of using the Consolidate feature for report aggregation.

Answer: It combines data from many sheets or workbooks into one summary without copy-paste. It matches by row and column labels and

offers Sum, Average, Count and more. With "Create links to source data" it can also refresh.

11.2 Using CONSOLIDATE Feature of Excel

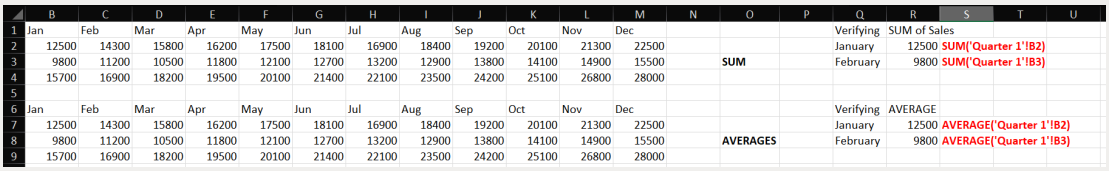
Task 1: Create a dataset with regional sales data across multiple sheets.

📸 **Answer:** The Quarter 1 to Quarter 4 sheets act as separate data sheets with the same row labels (Furniture, Office Supplies, Technology). Region sheets are built the same way.

Task 2: Use the CONSOLIDATE feature to combine the data into a summary sheet.

📸 **Answer:** The "Section 11 - Consolidation" sheet holds the summary table, built with Data → Consolidate (Sum, with four sheet references listed under All references).

Screenshot:



	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec				Verifying	SUM of Sales			
2	12500	14300	15800	16200	17500	18100	16900	18400	19200	20100	21300	22500				January	12500	SUM('Quarter 1'!B2)		
3	9800	11200	10500	11800	12100	12700	13200	12900	13800	14100	14900	15500		SUM		February	9800	SUM('Quarter 1'!B3)		
4	15700	16900	18200	19500	20100	21400	22100	23500	24200	25100	26800	28000								
5																				
6	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec				Verifying	AVERAGE			
7	12500	14300	15800	16200	17500	18100	16900	18400	19200	20100	21300	22500				January	12500	AVERAGE('Quarter 1'!B2)		
8	9800	11200	10500	11800	12100	12700	13200	12900	13800	14100	14900	15500		AVERAGES		February	9800	AVERAGE('Quarter 1'!B3)		
9	15700	16900	18200	19500	20100	21400	22100	23500	24200	25100	26800	28000								

Task 3: Experiment with consolidating data by categories.

📸 **Answer:** With "Left column" ticked, Excel matches rows by category name (Furniture, Office Supplies, Technology), so row order in each sheet does not matter.

Task 4: Apply the Consolidate feature to a different dataset for practice.

📸 **Answer:** Repeat the same steps on another set of sheets, for example sales by region per month: choose Sum, add each sheet's range and tick Top row and Left column.

Task 5: Explain how the CONSOLIDATE feature improves data management.

📸 **Answer:** It keeps one summary in sync with many sources, saves time, reduces manual errors and standardises reporting. Source data stays untouched.

11.3 Using SUBTOTAL Feature of Excel Using One Criterion

Task 1: Create a dataset with sales data categorized by product type.

📸 **Answer:** Table with Product Type, Product and Sales: Furniture (Chair 12,000; Table 18,000; Desk 14,000), Office Supplies (Paper 8,000; Binders 6,500; Storage 9,000) and Technology (Laptop 25,000; Monitor 15,000; Keyboard 5,000).

Task 2: Use the SUBTOTAL feature to calculate the total sales for each product type.

📸 **Answer:** `=SUBTOTAL(9,C2:C10)` gives **112,500** for all visible rows. Filtering by product type (for example Furniture) gives that type's total (44,000).

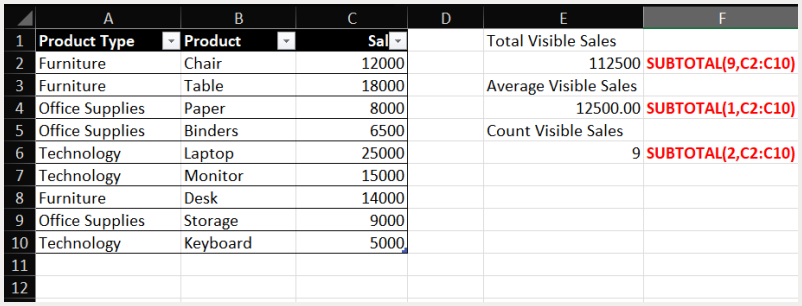
Task 3: Apply different SUBTOTAL functions (e.g., Average, Count) for analysis.

📸 **Answer:** `=SUBTOTAL(1,C2:C10)` = 12,500.00 (average) and `=SUBTOTAL(2,C2:C10)` = 9 (count of visible rows).

Task 4: Use filters in combination with the SUBTOTAL function to refine the data.

Answer: Use the filter drop-down on Product Type. The SUBTOTAL results (Total, Average, Count Visible Sales) recalculate for the visible rows only. A normal SUM would not change.

Screenshot:



	A	B	C	D	E	F
1	Product Type	Product	Sal		Total Visible Sales	
2	Furniture	Chair	12000		112500	SUBTOTAL(9,C2:C10)
3	Furniture	Table	18000		Average Visible Sales	
4	Office Supplies	Paper	8000		12500.00	SUBTOTAL(1,C2:C10)
5	Office Supplies	Binders	6500		Count Visible Sales	
6	Technology	Laptop	25000		9	SUBTOTAL(2,C2:C10)
7	Technology	Monitor	15000			
8	Furniture	Desk	14000			
9	Office Supplies	Storage	9000			
10	Technology	Keyboard	5000			
11						
12						

Task 5: Explain the importance of the SUBTOTAL feature in detailed data analysis.

Answer: SUBTOTAL ignores filtered-out rows and does not double-count other subtotals. It lets totals, averages and counts follow whatever the user filters, which suits dynamic report.

Section 12: Basics of Macros

12.1 What is VBA Macros?

Task 1: Research and write a brief explanation of what VBA Macros are.

Answer: VBA (Visual Basic for Applications) is the programming language built into Office. A macro is a recorded or written set of VBA instructions that automates repetitive tasks in Excel, such as formatting, copying data or creating reports.

Task 2: List the advantages and disadvantages of using VBA Macros.



Answer: Advantages: saves time, reduces errors, automates repetitive work, can be customised. Disadvantages: needs the .xlsm format, security risks, can break if the layout changes, needs VBA knowledge, and does not run in Excel Online.

Task 3: Identify scenarios where VBA Macros can be particularly useful.



Answer: Formatting headers or tables, adding borders, cleaning and merging data, generating monthly reports, sending emails, refreshing pivots and any task repeated daily or weekly.

Task 4: Explain the difference between recording a macro and writing VBA code.



Answer: Recording captures your actions and generates VBA automatically (quick, but verbose and limited, with no loops or conditions). Writing code gives full control, such as loops, IF logic, input boxes and error handling, and is cleaner and more flexible.

Task 5: Describe the security considerations when using VBA Macros.



Answer: Macros can carry malware, so only enable them in trusted files. Use Trust Center settings (disable with notification), keep files from unknown senders disabled, digitally sign macros and save as .xlsm so that macros are visible.

12.2 How to Enable Developer Tab in Excel?

Task 1: Enable the Developer tab in Excel by customizing the ribbon.

📸 **Answer:** File → Options → Customize Ribbon → tick Developer in the right-hand list → OK. The Developer tab now appears in the ribbon.

Task 2: Explore the different options available in the Developer tab.

📸 **Answer:** Groups: Code (Visual Basic, Macros, Record Macro, Use Relative References, Macro Security), Add-ins, Controls (Insert, Design Mode, Properties, View Code, Run Dialog) and XML (Source, Import, Export)

Task 3: Write a step-by-step guide on how to enable and use the Developer tab.

📸 **Answer:** 1) File → Options. 2) Customize Ribbon. 3) Tick Developer. 4) OK. 5) Open Developer → Record Macro to record, Macros to run, and Visual Basic (Alt+F11) to edit code. 6) Save the file as .xlsm.

Task 4: Explain the importance of the Developer tab for advanced Excel users.

📸 **Answer:** It gives access to macros, VBA, form controls (buttons, checkboxes) and XML tools. This enables automation and interactive dashboards.

Task 5: List the key features accessible through the Developer tab.

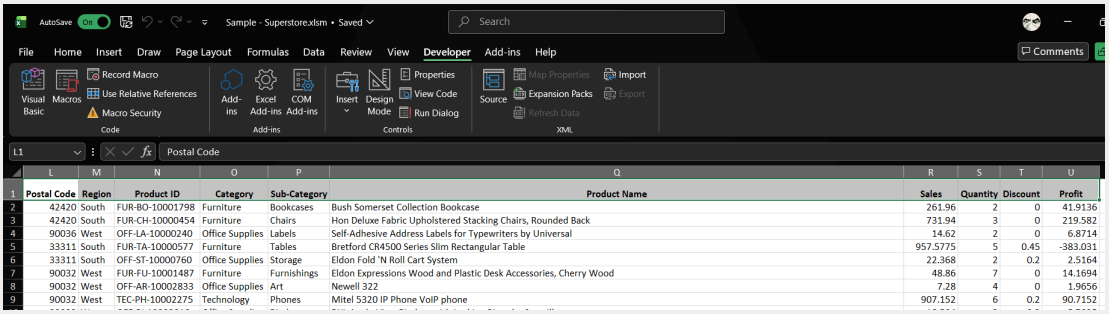
📸 **Answer:** Visual Basic editor, Macros, Record Macro, Relative References, Macro Security, Add-ins and COM Add-ins, form and ActiveX controls, Design Mode, View Code, Run Dialog, and XML Source, Import and Export.

12.3 Creating, Running and Saving a Macro

Task 1: Record a simple macro to automate a repetitive task (e.g., formatting a table).

Answer: Recorded the macro `FormatHeaders`. It makes the selected header row bold, centre-aligned, with no wrap text and no merge. Developer → Record Macro → perform the steps → Stop Recording.

Screenshot:



Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582
90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714
33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164
90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694
90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656
90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152

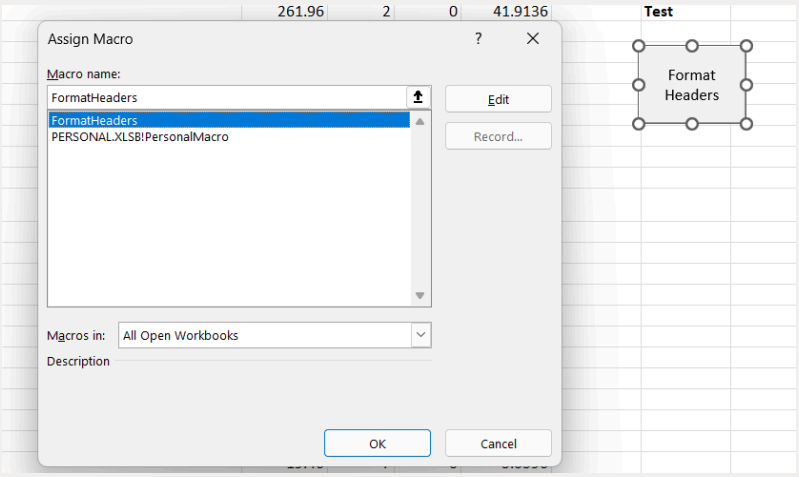
Task 2: Run the recorded macro and observe its effects.

Answer: Select the header row and run FormatHeaders (Developer → Macros → Run). The headers become bold and centred instantly.

Task 3: Save the macro in the Personal Macro Workbook for future use.

Answer: In Record Macro: name PersonalMacro, shortcut Ctrl+b, Store macro in: Personal Macro Workbook. It is saved in PERSONAL.XLSB and available in every workbook.

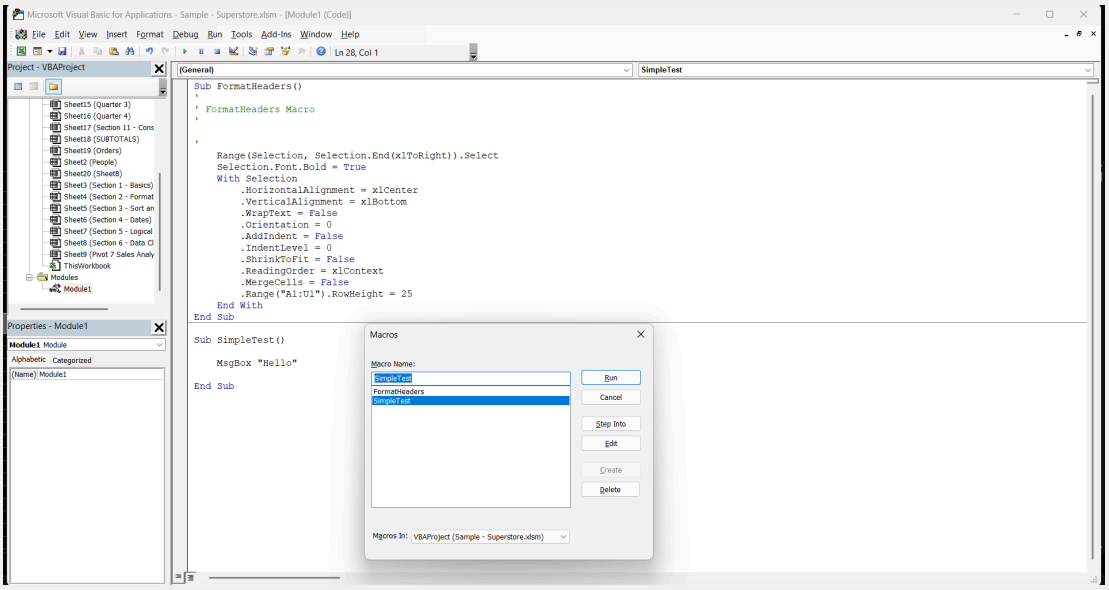
Screenshot:



Task 4: Edit the recorded macro to add or change functionality.

Answer: In the VBA editor the FormatHeaders macro was edited to add `.Range("A1:U1").RowHeight = 25`, which sets the header row height to 25.

Screenshot:



Task 5: Explain the process and benefits of recording and saving macros.

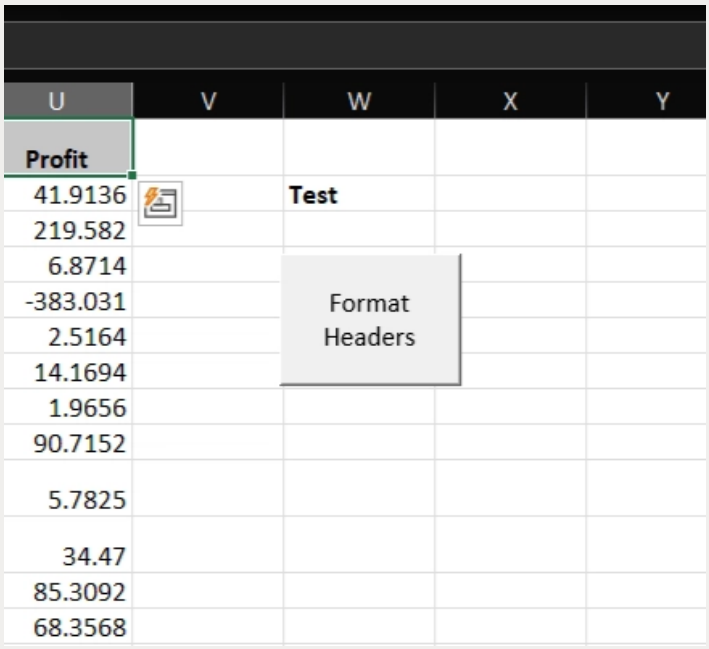
📸 **Answer:** Process: Record Macro → name, shortcut and storage location → perform the actions → Stop Recording → save as .xlsm. Benefits: saves time, ensures consistency, is reusable (especially in the Personal Macro Workbook) and needs no coding skills.

12.4 Running a Macro in Different Ways

Task 1: Assign a macro to a button on the worksheet and run it.

📸 **Answer:** Developer → Insert → Button → draw it → Assign Macro dialog → choose FormatHeaders → OK. The "Format Headers" button now runs the macro with a click.

Screenshot:



Task 2: Run a macro using a keyboard shortcut.

Answer: The macro PersonalMacro was given the shortcut Ctrl+b when recording. Press Ctrl+b to run it.

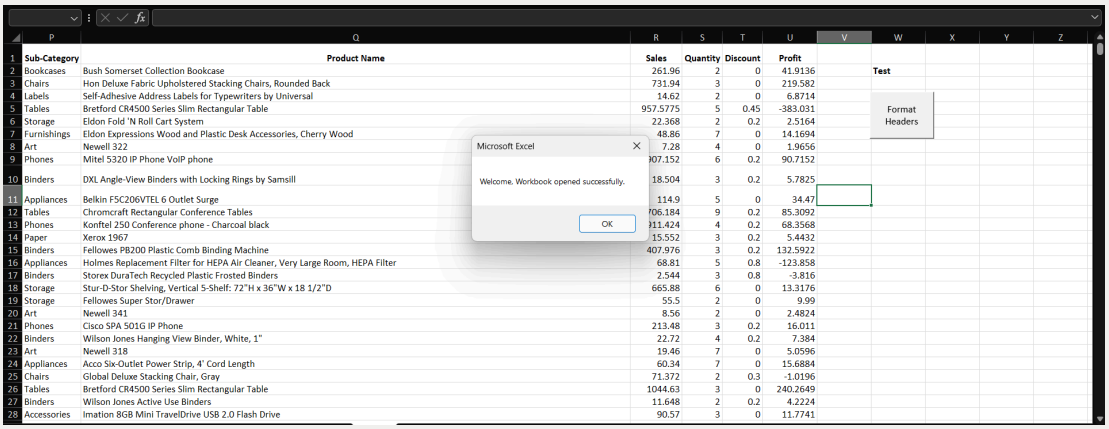
Task 3: Execute a macro from the Developer tab.

Answer: Developer → Macros opens the list (AddBorders, FormatHeaders, PERSONAL.XLSB!PersonalMacro, SimpleTest). Pick one and click Run.

Task 4: Run a macro automatically when a workbook is opened.

Answer: A Workbook_Open event in ThisWorkbook runs on opening and displays "Welcome, Workbook opened successfully."

Screenshot:



	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
2	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
3	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582
4	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714
5	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
6	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164
7	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	48.86	7	0	14.1694
8	Art	Newell 322	7.28	4	0	1.9656
9	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152
10	Binders	DXL Angle-View Binders with Locking Rings by Samsill	18.504	3	0.2	5.7825
11	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47
12	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
13	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568
14	Paper	Xerox 1967	15.552	3	0.2	5.4432
15	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922
16	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	5	0.8	-123.858
17	Binders	Storex DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.816
18	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	6	0	13.3176
19	Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99
20	Art	Newell 341	8.56	2	0	2.4824
21	Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011
22	Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384
23	Art	Newell 318	19.46	7	0	5.0596
24	Appliances	Acco Six-Outlet Power Strip, 4' Cord Length	60.34	7	0	15.6884
25	Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0196
26	Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649
27	Binders	Wilson Jones Active Use Binders	11.648	2	0.2	4.2224
28	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741

Task 5: Explain the different methods of running macros and their use cases.

Answer: Developer → Macros → Run (general use); keyboard shortcut (quick personal tasks); button on the sheet (for non-technical users);

Workbook_Open event (automatic set-up or welcome messages); and Run from the VBA editor (testing).

12.5 Understanding VBA Workspace

Task 1: Open the Visual Basic for Applications (VBA) editor.

📸 **Answer:** Developer → Visual Basic, or press Alt+F11. The editor opens with the project "VBAProject (Sample - Superstore.xlsm)".

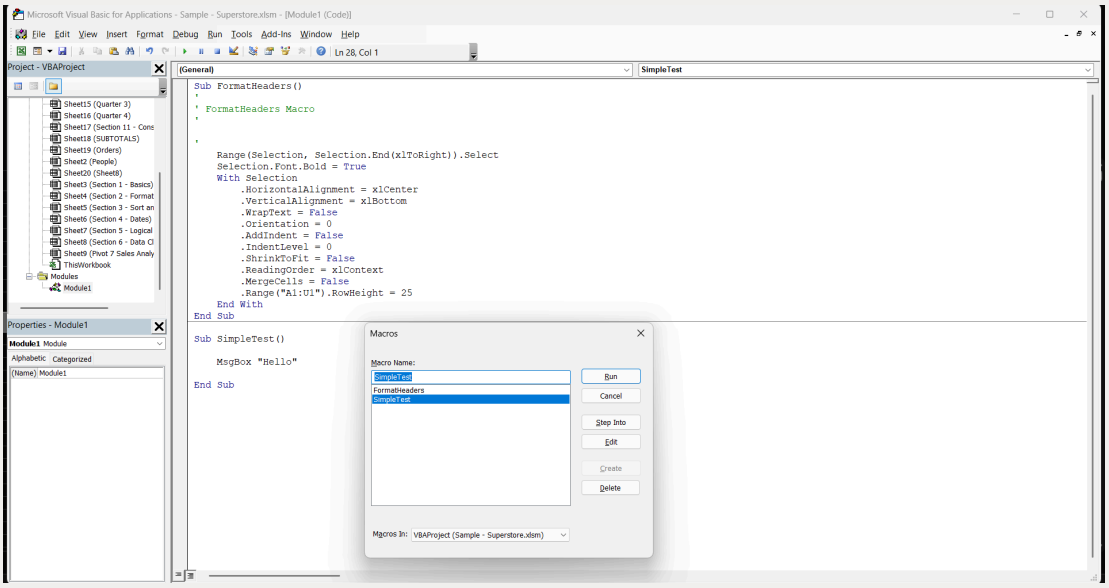
Task 2: Explore the different components of the VBA workspace (e.g., Project Explorer, Code Window).

📸 **Answer:** Project Explorer (top left: workbook, sheets, ThisWorkbook and Modules → Module1), Properties window (bottom left, shows Module1's name), Code window (right, holds the macro code), plus the menu bar and toolbar.

Task 3: Write a simple VBA script in the editor.

📸 **Answer:** `Sub SimpleTest()` with `MsgBox "Hello"` and `End Sub`, written in Module1 below FormatHeaders.

Screenshot:



Task 4: Run the VBA script and observe its effects.

Answer: Macros dialog → SimpleTest → Run. A message box titled Microsoft Excel appears showing "Hello".

Screenshot:

	P	Q	R	S	T	U
1	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
2	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
3	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	731.94	3	0	219.582
4	Labels	Self-Adhesive Address Labels for Typewriters by Universal	14.62	2	0	6.8714
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16	Appliances	Holmes Replacement Filter for HEPA Air Cleaner, Very Large Room, HEPA Filter	68.81	5	0.8	-123.858
17	Binders	Storex DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.816
18	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W x 18 1/2"D	665.88	6	0	13.3176
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22	Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384
23	Art	Newell 318	19.46	7	0	5.0596
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25	Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0196
26	Tables	Bretford CR4500 Series Slim Rectangular Table	1044.63	3	0	240.2649
27	Binders	Wilson Jones Active Use Binders	11.648	2	0.2	4.2224
28	Accessories	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741

Task 5: Explain the layout and functionalities of the VBA workspace.



Answer: The Project Explorer manages workbooks, sheets and modules. The Code window is where you write and edit macros. The Properties window shows object settings. The toolbar runs, pauses and stops code, and the Immediate and Watch windows (View menu) help with debugging.

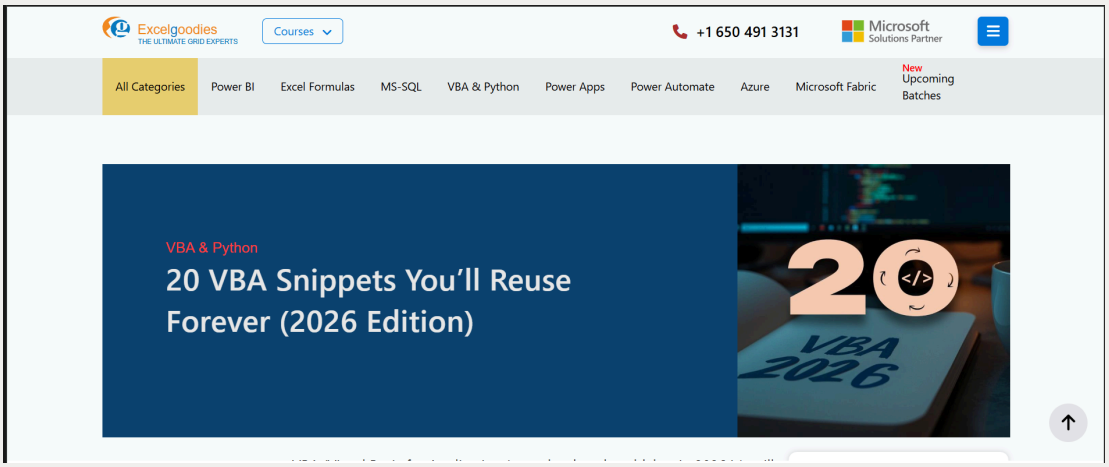
12.6 How to Get VBA Codes?

Task 1: Research online resources for finding VBA code snippets.



Answer: Searched online and found ExcelGoodies' "20 VBA Snippets You'll Reuse Forever (2026 Edition)". Other sources: Microsoft Learn, Stack Overflow, MrExcel, Chandoo and Contextures.

Screenshot:



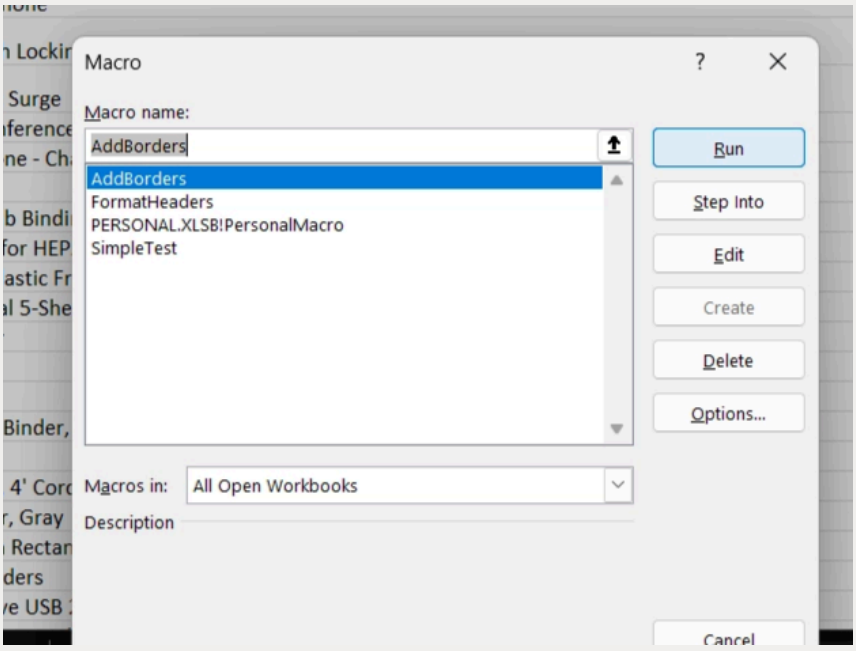
Task 2: Identify reputable sources for VBA code examples.

Answer: Microsoft Learn (official documentation), ExcelGoodies, Stack Overflow, MrExcel, Chandoo and Contextures. Prefer sources with explanations, active communities and recent updates.

Task 3: Copy a VBA code snippet from an online resource and customize it for your needs.

Answer: Created the macro `AddBorders` with `Selection.Borders.LineStyle = xlContinuous`. It puts a continuous border around whatever range is selected.

Screenshot:



```
Sub AddBorders()

    Selection.Borders.LineStyle = xlContinuous

End Sub
```

Task 4: Test the customized VBA code in your workbook.

📸 **Answer:** Selected the data range, opened Developer → Macros, chose AddBorders and clicked Run. Borders were applied to the selection. The macro appears in the list next to FormatHeaders and SimpleTest.

Task 5: Explain the process of finding, copying, and using VBA code snippets.

📸 **Answer:** Search a trusted source → copy the code → open Alt+F11 → Insert → Module → paste → adjust ranges and names → save as .xlsm → run from Developer → Macros or a button. Always test on a copy first and never run code you do not understand.